Final Lab Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	40
Understanding/Analysis	10					
Implementation	15					
Accuracy	10					
Task Efficiency	5					
Total obtained mark						
Comments						

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FarmFusion – Smart Agro System

COMPLETE PROJECT REPORT

SE231 – System Analysis & Design Capstone Project

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Abstract

FarmFusion – Smart Agro System is an integrated digital farm-management platform designed to connect core agricultural operations in one system. The project brings together user registration and role-based access, animal health monitoring, automated inventory management, feeding and nutrition management, crop harvesting, dairy product management, dairy product wholesale, sales and billing, expense tracking, and real-time weather monitoring with alerts.

The system is designed around the operational roles identified in the project models, including Admin, Staff, Veterinarian, Farmer, Wholesaler/Customer, and automated environmental devices. The supplied use case, activity, sequence, state, class, ER, and DFD models were used as the main design basis for this report. The project emphasizes centralized data handling, traceability, timely alerts, and a simple role-oriented user experience.

The report follows the structure of the supplied sample project report: introduction, literature review and requirements, behavioral analysis, methodology, system design, prototypes, implementation and testing, learning outcomes, project reflection, conclusion, next steps, and references.

Chapter 1: Introduction

1.1 Background & Overview

Agricultural operations often involve several connected activities: animal care, feeding, inventory, crop production, dairy processing, sales, financial recording, and weather-dependent decisions. When these activities are handled separately, information can become fragmented and operational decisions may depend heavily on manual records. A smart agro system can reduce this fragmentation by providing a centralized platform where operational data can be recorded, monitored and shared with authorized users.

FarmFusion is proposed as a centralized smart agro-management environment. Its design reflects the project's supplied diagrams, where operational processes are connected through a common system and database. The system also considers automated environmental data and alert generation, allowing users to respond to important conditions without relying only on manual observation.

1.2 Problem Statement

- Farm operations may use disconnected records for animals, inventory, crops, dairy products and finance.
- Manual animal-health records can make it difficult to maintain a consistent history of diagnosis, treatment, newborn monitoring and pregnancy tracking.
- Inventory shortages of feed, medicines, vaccines and other essential items may be noticed late.
- Crop harvesting requires coordination of maturity checks, resources, yield, quality and dispatch.
- Dairy production and wholesale activities require consistent product quantities, quotations, orders and delivery reconciliation.
- Sales, billing and expense records need reliable verification so that financial reports remain consistent.
- Weather and environmental changes can affect farm operations, but without timely alerts users may react too late.
- Different actors require different permissions, dashboards and responsibilities.

1.3 Project Objectives

- To provide a centralized smart agro management platform for major farm operations.
- To support secure registration, login and role-based access.
- To maintain structured animal health records and monitoring information.
- To automate inventory monitoring and low-stock alert generation.
- To manage feeding schedules, food options, staff assignments and feeding confirmations.
- To support crop harvest planning, maturity checking, resource planning, yield collection, quality inspection and dispatch.
- To manage dairy products and wholesale order workflows.
- To record sales, invoices, payments, expenses and financial reports.
- To receive environmental data and generate real-time or forecast-based weather alerts.
- To provide traceable and consistent data flows across the system.

1.4 Scope and Limitations

Scope

- User registration, verification and login.
- Role-based dashboards for Admin, Staff, Veterinarian and Farmer.
- Animal health monitoring and health-record management.

- Automated inventory and low-stock monitoring.
- Feeding and nutrition planning and session confirmation.
- Crop harvesting planning and harvest-record management.
- Dairy product production and stock management.
- Dairy product wholesale request, quotation, negotiation and delivery reconciliation.
- Sales order, invoice, payment, expense and financial-report management.
- Weather/environmental data monitoring and risk alerts.
- Centralized database and system activity records.

Limitations

- The quality of automated weather information depends on the availability and accuracy of the connected data source.
- Automated camera and sensor features require compatible hardware and network connectivity.
- Financial calculations depend on the correctness of user-entered transaction data.
- Advanced predictive analytics may require additional historical datasets and machine-learning models.
- The report documents the academic project scope; production deployment would require additional security, infrastructure and operational validation.

Chapter 2: Literature Review

Smart farming combines digital technologies with agricultural operations to improve resource management and decision making. FAO describes smart farming in terms of affordable technologies, efficient resource management and digital technologies that help farmers produce more with fewer inputs. □cite□turn0search15□

Digital livestock management also increasingly emphasizes electronic identification, precision feeding, environmental sensing, monitoring and early warning. These ideas are consistent with FarmFusion's animal-health, feeding, inventory and alert modules. □cite□turn0search48□

From a software-engineering perspective, FarmFusion requires both functional correctness and quality attributes such as usability, security, reliability and maintainability. ISO/IEC 25010:2023 provides a current product-quality model that can be used as a reference when defining and evaluating software quality requirements. □cite□turn0search0□

2.1 Feasibility Analysis

Technical Feasibility

The project is technically feasible because the required components can be implemented using standard web technologies, a database layer, diagramming tools and optional device/API integrations. The supplied class and ER models demonstrate that the core data structures can be represented systematically, while the sequence and activity models demonstrate implementable request and workflow paths.

Operational Feasibility

The proposed system separates responsibilities by role and presents operational tasks through dedicated modules. This supports adoption because users can focus on the functions relevant to their work instead of interacting with every system feature.

Economic Feasibility

For an academic prototype, the system can be implemented using widely available development and modeling tools. Production deployment would require additional costs for hosting, device installation, notification services, backup, maintenance and security monitoring.

2.2 Requirements Specification

2.2.1 User Registration & Authentication

Requirement Group	FR-REG/AUTH
Description	The system shall allow authorized roles to register, verify accounts, log in and access only permitted dashboards.
Stakeholders	Admin, Staff, Veterinarian, Farmer and/or relevant external party as defined by the module.

The system [FR-REG/AUTH] shall implement the described capability while maintaining role-based access, validation and traceable records.

2.2.2 Animal Health Monitoring

Requirement Group	FR-ANH
Description	The system shall record animal health, newborn movement, pregnancy monitoring, treatments and essential medicine/vaccine status.

Stakeholders	Admin, Staff, Veterinarian, Farmer and/or relevant external party as defined by the module.

The system [FR-ANH] shall implement the described capability while maintaining role-based access, validation and traceable records.

2.2.3 Automated Inventory Management

Requirement Group	FR-INV
Description	The system shall maintain item quantities, reorder levels and low-stock alerts.
Stakeholders	Admin, Staff, Veterinarian, Farmer and/or relevant external party as defined by the module.

The system [FR-INV] shall implement the described capability while maintaining role-based access, validation and traceable records.

2.2.4 Feeding & Nutrition Management

Requirement Group	FR-FED
Description	The system shall support feed tracking, meal schedules, feed/staff selection, feeding orders and confirmation.
Stakeholders	Admin, Staff, Veterinarian, Farmer and/or relevant external party as defined by the module.

The system [FR-FED] shall implement the described capability while maintaining role-based access, validation and traceable records.

2.2.5 Crop Harvesting Management

Requirement Group	FR-CROP
Description	The system shall support harvest planning, maturity checking, resource planning, yield collection, quality inspection and dispatch.
Stakeholders	Admin, Staff, Veterinarian, Farmer and/or relevant external party as defined by the module.

The system [FR-CROP] shall implement the described capability while maintaining role-based access, validation and traceable records.

2.2.6 Dairy Product Management

Requirement Group	FR-DAIRY
Description	The system shall record dairy product entries, damages, quantities and stock updates.
Stakeholders	Admin, Staff, Veterinarian, Farmer and/or relevant external party as defined by the module.

The system [FR-DAIRY] shall implement the described capability while maintaining role-based access, validation and traceable records.

2.2.7 Dairy Product Wholesale Management

Requirement Group	FR-WHOLE
Description	The system shall support product requests, availability, quotations, negotiation, confirmation/rejection and delivery reconciliation.
Stakeholders	Admin, Staff, Veterinarian, Farmer and/or relevant external party as defined by the module.

The system [FR-WHOLE] shall implement the described capability while maintaining role-based access, validation and traceable records.

2.2.8 Sales, Billing & Expense Tracking

Requirement Group	FR-FIN
Description	The system shall support sales orders, invoice generation, payment collection, expense recording, verification and financial reporting.
Stakeholders	Admin, Staff, Veterinarian, Farmer and/or relevant external party as defined by the module.

The system [FR-FIN] shall implement the described capability while maintaining role-based access, validation and traceable records.

2.2.9 Weather Monitoring & Alert Management

Requirement Group	FR-WEATHER
Description	The system shall receive environmental data, analyze conditions and generate timely alerts.
Stakeholders	Admin, Staff, Veterinarian, Farmer and/or relevant external party as defined by the module.

The system [FR-WEATHER] shall implement the described capability while maintaining role-based access, validation and traceable records.

Non-Functional Requirements

Quality Attribute	**Requirement**
Performance	Common dashboard actions should respond promptly; alerts and operational confirmations should be delivered without unnecessary delay.
Security & Privacy	Authentication, authorization, password protection and access controls shall prevent unauthorized access to operational and financial data.
Data Integrity	Transactions and records shall be validated before saving; related records shall remain consistent.
Reliability	The system should preserve valid data and handle service/device failures without silently corrupting records.
Usability	Interfaces should be responsive, readable and organized around role-specific tasks.
Scalability	The design should allow additional farms, users, devices and operational records without redesigning the entire system.
Maintainability	Modules should remain separated so that individual services and workflows can be updated with limited impact on unrelated functions.
Auditability	Important changes to health, inventory, finance and access records should be traceable through timestamps and user identity where applicable.

2.3 Behavioral Analysis & Requirements Modeling

2.3.1 Use Case Diagram

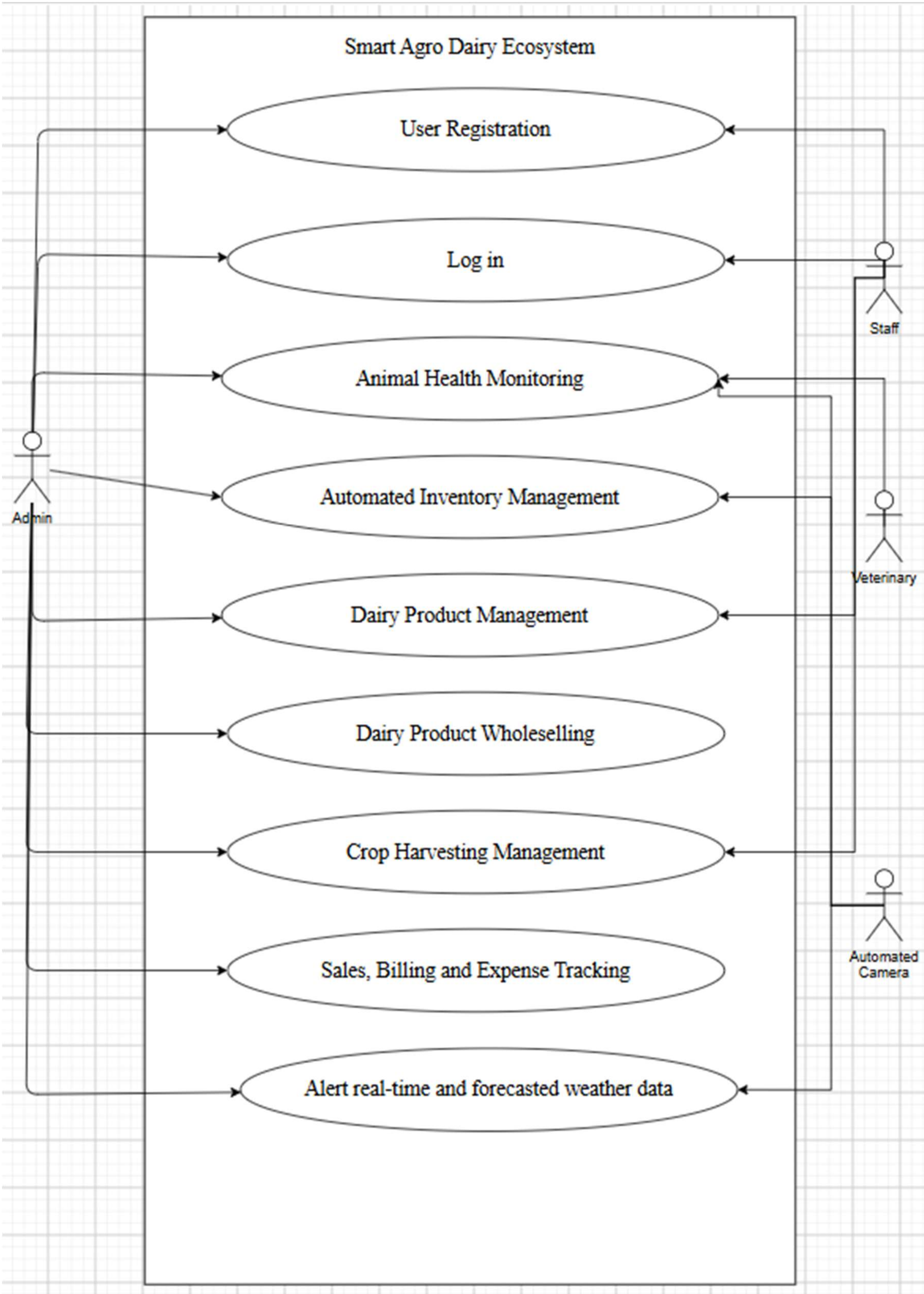

Figure 2.3.1: Use Case Diagram of the FarmFusion – Smart Agro System

The use case diagram identifies the principal system actors and the major services available within FarmFusion. It shows registration, login, animal health monitoring, automated inventory, dairy product management, dairy product wholesaling, crop harvesting, sales/billing/expense tracking and real-time or forecasted weather information as the core system functions.

2.3.2 Use Case Descriptions

Case Description-01: User Registration

Use Case	User Registration
Goal	A new authorized user successfully creates and verifies an account in FarmFusion.
Precondition	The user is on the Registration page and does not already have an active account.
Success End Condition	The submitted information is validated, the account is created, verification is completed, and the user can access the Login page.
Failed End Condition	The system rejects invalid/duplicate information or failed verification and does not activate the account.
Primary Actors	User (Admin/Staff/Veterinarian/Farmer)
Secondary Actors	Verification/Notification Service
Trigger	The user submits the registration form.
Description / Main Success Scenario	1. Open Registration page. 2. Enter required personal and role information. 3. Submit registration data. 4. System validates required fields and uniqueness. 5. System sends OTP/verification request. 6. User submits the OTP. 7. System verifies the OTP and creates the account. 8. System confirms successful registration and provides access to Login.
Alternative Flows	1. Invalid or missing data → show validation errors and request correction. 2. Existing email/username → show duplicate-account message. 3. Invalid/expired OTP → request a new OTP or retry verification.
Quality Requirements	Required fields must be validated; credentials must be protected; verification should be completed promptly.

Case Description-02: User Login

Use Case	User Login
Goal	An authorized user successfully logs into FarmFusion and reaches the appropriate role-based dashboard.
Precondition	The user has a verified account and is on the Login page.
Success End Condition	Credentials are validated and the system redirects the user to the dashboard permitted for the user's role.
Failed End Condition	Invalid credentials prevent access and an error message is displayed.
Primary Actors	User (Admin/Staff/Veterinarian/Farmer)
Secondary Actors	User Database
Trigger	The user submits a username/email and password.

Description / Main Success Scenario	1. Open Login page. 2. Enter registered credentials. 3. Submit Login. 4. System retrieves matching account data. 5. System validates credentials. 6. System checks the user's role and permissions. 7. System creates an authenticated session. 8. System redirects the user to the correct dashboard.
Alternative Flows	1. Invalid credentials → display error and allow retry. 2. Inactive/unverified account → deny access and request verification. 3. Unauthorized role access → deny the requested page and return to the permitted dashboard.
Quality Requirements	Authentication must be secure; protected pages must require a valid session; role-based access must be enforced.

Case Description-03: Animal Health Monitoring

Use Case	Animal Health Monitoring
Goal	Veterinary staff and authorized staff continuously maintain accurate animal-health information.
Precondition	The authorized user is logged in and animal records exist or can be created.
Success End Condition	Health records, newborn movement, pregnancy monitoring, medicines and vaccines are recorded and available for monitoring.
Failed End Condition	Invalid or incomplete health data is rejected and the user is asked to correct it.
Primary Actors	Veterinarian, Staff
Secondary Actors	Automated Camera, Inventory Database
Trigger	An authorized user records or requests animal-health information, or an automated event is received.
Description / Main Success Scenario	1. Open Animal Health Monitoring. 2. Select an animal or create a record. 3. Enter health status, diagnosis/treatment and relevant observations. 4. Record newborn movement and pregnancy-related monitoring when applicable. 5. System stores the health record. 6. System checks essential medicines/vaccines in stock. 7. System displays health information and relevant alerts.
Alternative Flows	1. Invalid animal/health data → show validation error. 2. Medicine/vaccine stock below threshold → generate alert for responsible staff/admin. 3. Camera data unavailable → retain the latest valid record and indicate unavailable sensor data.
Quality Requirements	Health data should be accurate, access-controlled, traceable and stored without unauthorized modification.

Case Description-04: Automated Inventory Management

Use Case	Automated Inventory Management
Goal	The system maintains inventory quantities and warns responsible users when stock is insufficient.

Precondition	Inventory items and reorder thresholds are configured.
Success End Condition	Stock is updated correctly and low-stock items generate alerts.
Failed End Condition	Invalid stock changes are rejected and the existing quantity remains unchanged.
Primary Actors	Admin, Staff
Secondary Actors	Database
Trigger	A user adds, updates or checks an inventory item.
Description / Main Success Scenario	1. Open Inventory Management. 2. Add or select an inventory item. 3. Enter/update quantity and reorder level. 4. System validates the values. 5. System saves the inventory record. 6. System compares stock with the threshold. 7. System generates a low-stock alert when required. 8. System displays the updated inventory status.
Alternative Flows	1. Negative/invalid quantity → reject the update. 2. Duplicate item → prompt the user to update the existing item. 3. Low stock → notify Admin/Staff for replenishment.
Quality Requirements	Inventory quantities must remain consistent; updates should be auditable; alerts should be timely.

Case Description-05: Dairy Product Management

Use Case	Dairy Product Management
Goal	Authorized users record and manage dairy product information and production stock.
Precondition	The user is authenticated and has dairy-management permission.
Success End Condition	A dairy product entry is recorded, quantities/damages are updated, and the product becomes available in inventory/sales workflows.
Failed End Condition	Invalid product or quantity information prevents saving.
Primary Actors	Staff, Admin
Secondary Actors	Database
Trigger	A user adds, edits or confirms a dairy product record.
Description / Main Success Scenario	1. Open Dairy Product Management. 2. Enter product details and quantity. 3. Record damage/wastage information when applicable. 4. Submit the entry. 5. System validates the data. 6. System stores the product record and updates stock. 7. System confirms the new entry.
Alternative Flows	1. Invalid quantity/product data → show validation error. 2. Duplicate/unfinished entry → allow correction before confirmation. 3. Database failure → do not confirm the entry and request retry.
Quality Requirements	Product quantity and production data should be consistent and traceable.

Case Description-06: Dairy Product Wholesale Management

Use Case	Dairy Product Wholesale Management
Goal	The system manages wholesale product requests from availability checking through delivery reconciliation.
Precondition	Products are available in the dairy inventory and a wholesale party is registered/authorized.
Success End Condition	The request is negotiated, confirmed or rejected, and delivery information is reconciled.
Failed End Condition	The request cannot proceed because products are unavailable, details are invalid, or the request is rejected.
Primary Actors	Wholesaler/Supplier, Admin
Secondary Actors	Dairy Product Inventory, Notification Service
Trigger	A wholesaler requests products with quantity and delivery requirements.
Description / Main Success Scenario	1. Submit product request with quantity and required time. 2. System checks and displays availability. 3. Request quotation. 4. Admin reviews and negotiates terms. 5. Admin confirms or rejects the request. 6. For confirmed requests, the system records the order and delivery details. 7. System reconciles delivered quantity/status with the order.
Alternative Flows	1. Insufficient availability → inform requester and suggest available quantity. 2. Quotation rejected → close the request without confirmation. 3. Delivery mismatch → flag discrepancy for reconciliation.
Quality Requirements	Order and quotation records must be consistent, transparent and traceable.

Case Description-07: Crop Harvesting Management

Use Case	Crop Harvesting Management
Goal	The system manages crop harvesting from planning through dispatch.
Precondition	Crop records exist and harvesting is planned or due.
Success End Condition	Harvest is executed, yield and quality are recorded, and the harvested output is dispatched/updated in records.
Failed End Condition	Harvest cannot proceed when maturity or required resources are insufficient.
Primary Actors	Staff, Admin, Farmer
Secondary Actors	Weather Data, Crop Database
Trigger	An authorized user starts a harvest plan or a crop reaches harvest readiness.
Description / Main Success Scenario	1. Open Harvest Planning. 2. Select the field/crop section. 3. Check crop maturity. 4. Plan required resources. 5. Execute harvest. 6. Record yield. 7. Inspect quality. 8. Dispatch harvested produce and update the harvest record.
Alternative Flows	1. Crop not mature → keep harvest pending and reschedule.

	2. Insufficient resources → return to resource planning. 3. Poor quality → flag the batch for review before dispatch.
Quality Requirements	Harvest data should be accurate, time-stamped and linked to the relevant crop/field.

Case Description-08: Sales, Billing & Expense Tracking

Use Case	Sales, Billing & Expense Tracking
Goal	Admin records sales, billing and expenses and obtains reliable financial information.
Precondition	Admin is authenticated and financial data entry is available.
Success End Condition	Sales orders, invoices, payments and expenses are recorded and the financial dashboard/report is updated.
Failed End Condition	Invalid financial data is rejected or flagged for discrepancy.
Primary Actors	Admin
Secondary Actors	Database
Trigger	Admin opens the Sales, Billing & Expense Tracking module.
Description / Main Success Scenario	1. Create a sales order. 2. Generate an invoice. 3. Record payment. 4. Record expense. 5. Categorize the expense. 6. System verifies the financial data. 7. System calculates/update financial metrics. 8. System updates the financial dashboard and reports.
Alternative Flows	1. Invalid amount/date → show validation error. 2. Verification failure → flag discrepancy and request correction. 3. Payment not confirmed → keep invoice/payment status pending.
Quality Requirements	Financial records must be accurate, consistent, auditable and access-controlled.

Case Description-09: Weather Monitoring & Alert Management

Use Case	Weather Monitoring & Alert Management
Goal	The system monitors environmental conditions and provides timely weather-risk alerts.
Precondition	Environmental data source is available and alert thresholds are configured.
Success End Condition	Current/forecast data is received, analyzed and relevant alerts are sent to responsible users.
Failed End Condition	If environmental data is unavailable, the system records the failure and avoids issuing unsupported alerts.
Primary Actors	Automated Camera/Weather Device, System
Secondary Actors	Admin, Farmer
Trigger	A scheduled sensor/weather-data update is received.
Description / Main Success Scenario	1. Receive real-time sensor/environmental data. 2. Retrieve forecast information when available. 3. Check weather conditions against configured risk thresholds. 4. Detect immediate or forecasted risk. 5. Generate an alert when risk is detected. 6. Send SMS/push/in-app notification to responsible users.

	7. Store the weather event and alert record.
Alternative Flows	1. No risk → store/update weather data without sending a warning. 2. Sensor unavailable → record data-source failure and use the latest valid information where appropriate. 3. Severe condition → escalate alert priority to Admin and Farmer.
Quality Requirements	Alerts should be timely, relevant, reliable and based on validated environmental data.

2.3.3 Data Flow Diagram (DFD)

The DFD was normalized from the project's sequence, activity, class and ER information. It separates external actors, processing functions and persistent data stores and shows the movement of operational data through FarmFusion.

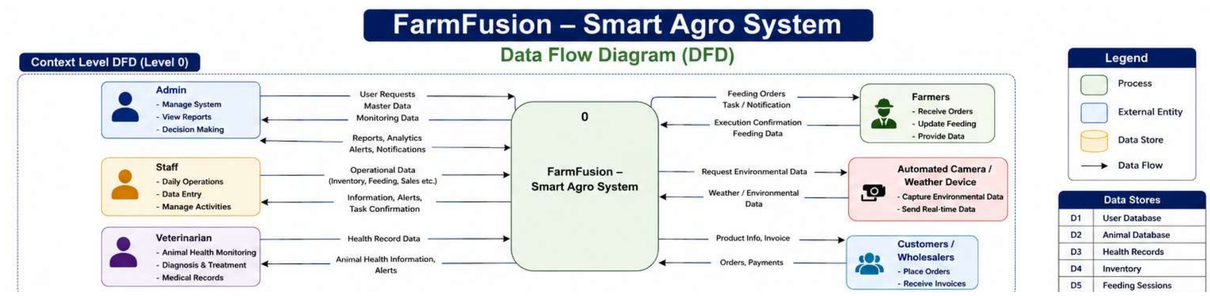

Figure 2.3.2: Level 0 DFD of the FarmFusion – Smart Agro System

Level 0 treats FarmFusion as a single process and shows the principal external entities: Admin, Staff, Veterinarian, Farmer, Customers/Wholesalers and automated environmental devices.

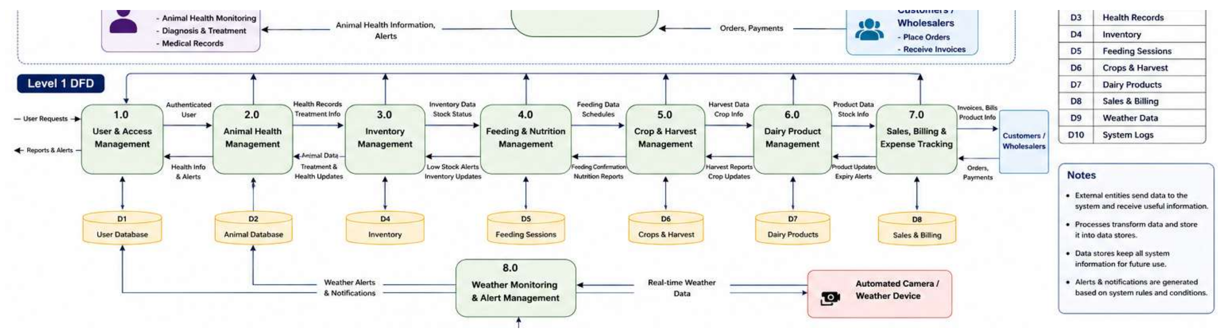

Figure 2.3.3: Level 1 DFD of the FarmFusion – Smart Agro System

Level 1 decomposes the system into major operational processes and their related data stores, including user/access, animal health, inventory, feeding, crops/harvest, dairy products, sales/billing and weather management.

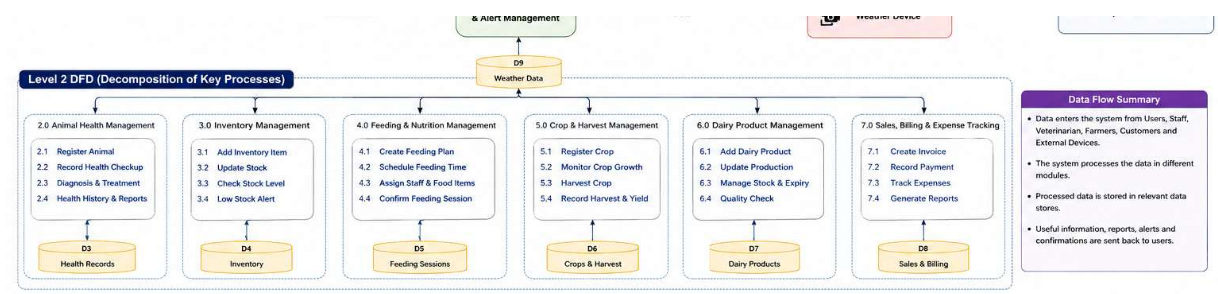

Figure 2.3.4: Level 2 DFD – Decomposition of Key FarmFusion Processes

2.4 Methodology

2.4.1 SDLC Model Selection (Agile Architecture)

Agile is suitable for FarmFusion because the project contains several modules that can be developed and refined iteratively. The Agile principles emphasize early and continuous delivery, welcoming changing requirements and frequent delivery of working software. cite turn0search2

- Requirements can be refined as individual farm modules are understood.
- Each sprint can focus on a small group of functions such as authentication, animal health or inventory.
- Testing can be performed continuously as modules are integrated.
- Feedback can be incorporated before the entire system is finalized.
- Diagram consistency can be checked after each major change.

2.4.2 Work Breakdown Structure (WBS) & Development Pipeline

Phase	Timeline	Key Activities
Phase 1: Inception & Planning	Weeks 1–2	Problem identification, stakeholder analysis, project scope and objectives.
Phase 2: System Analysis & Design	Weeks 3–4	Requirements, use cases, class/ER/DFD/UML diagrams and database planning.
Phase 3: UI/UX & Core Development	Weeks 5–8	Responsive dashboards and core operational modules.
Phase 4: Integration & Automation	Weeks 9–11	Database integration, alerts, device/API integration and module integration.
Phase 5: Testing & Final Review	Weeks 12–14	Functional testing, integration testing, documentation and presentation.

2.4.3 Development Environment & Tools Stack

- Frontend: HTML5, CSS3 and JavaScript for the responsive web prototype.
- Backend/Application Logic: a modular server-side application layer suitable for authentication, APIs and business logic.
- Database: a structured data layer representing the entities and records shown in the ER and class diagrams.
- Automation/Device Integration: automated camera/environmental data sources and notification services.
- Modeling: diagrams.net/Draw.io or equivalent UML/diagramming tools.
- Documentation: Microsoft Word-compatible report generation and structured tables/figures.

Chapter 3: System Design

3.1 Structural & Architectural Design

3.1.1 Block Diagram

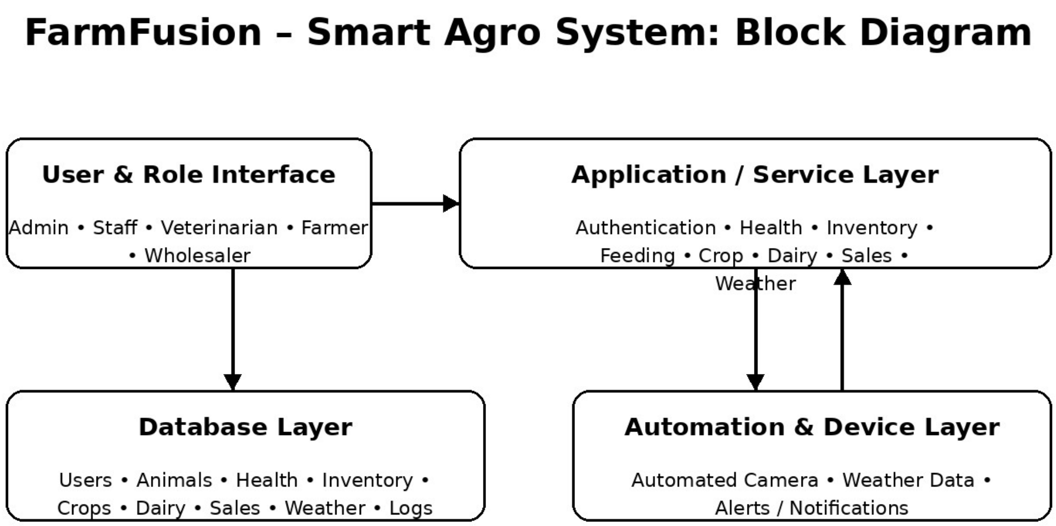

Figure 3.1.1: Block Diagram of the FarmFusion – Smart Agro System

The block diagram presents the system as four cooperating layers: the user and role interface, the application/service layer, the database layer, and the automation/device layer. This structure provides a clear separation between presentation, business processing, persistent data and external environmental inputs.

3.1.2 Class Diagram

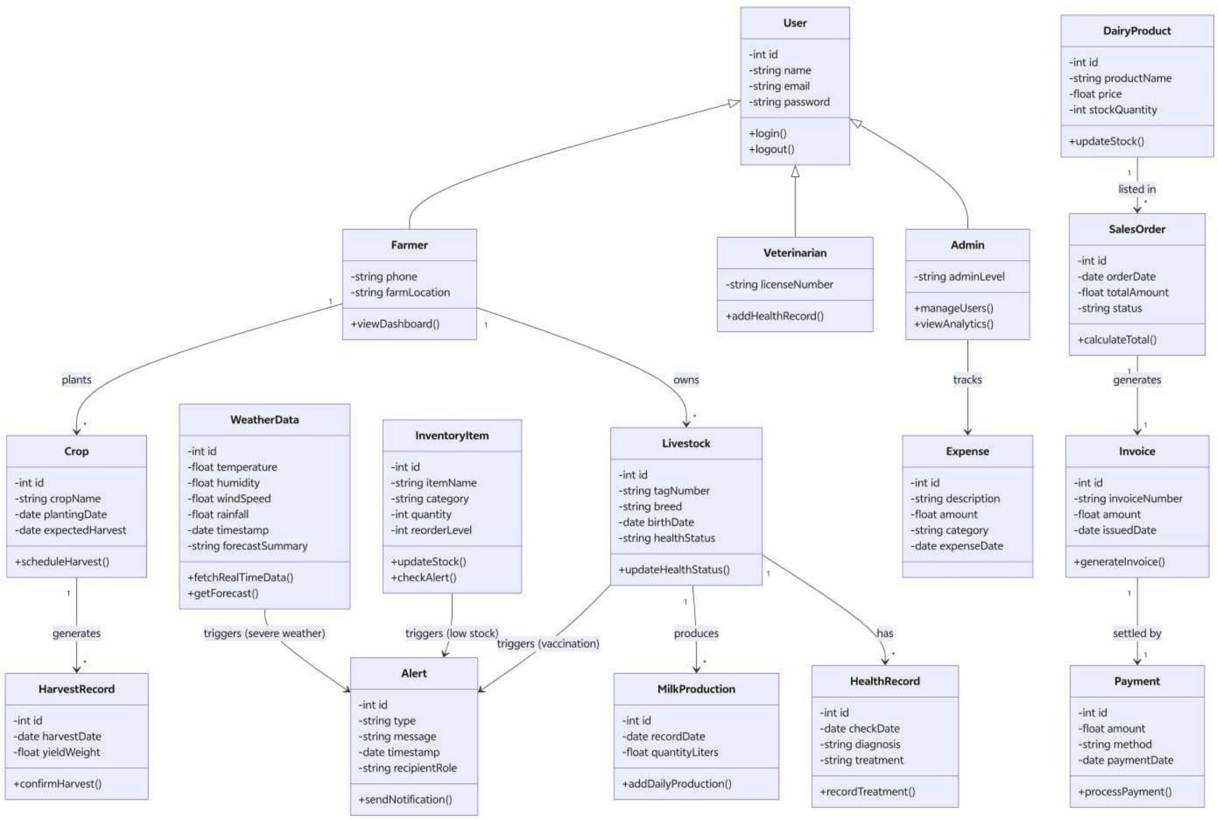

Figure 3.1.2: Class Diagram of the FarmFusion – Smart Agro System

The class diagram represents the static structure of FarmFusion. It includes User and role-specialized classes such as Farmer, Veterinarian and Admin, together with operational classes such as Livestock, HealthRecord, MilkProduction, InventoryItem, Crop, HarvestRecord, DairyProduct, SalesOrder, Invoice, Payment, Expense, WeatherData and Alert. Attributes and methods represent the information and operations required by the system.

3.1.3 Entity-Relationship Diagram (ERD)

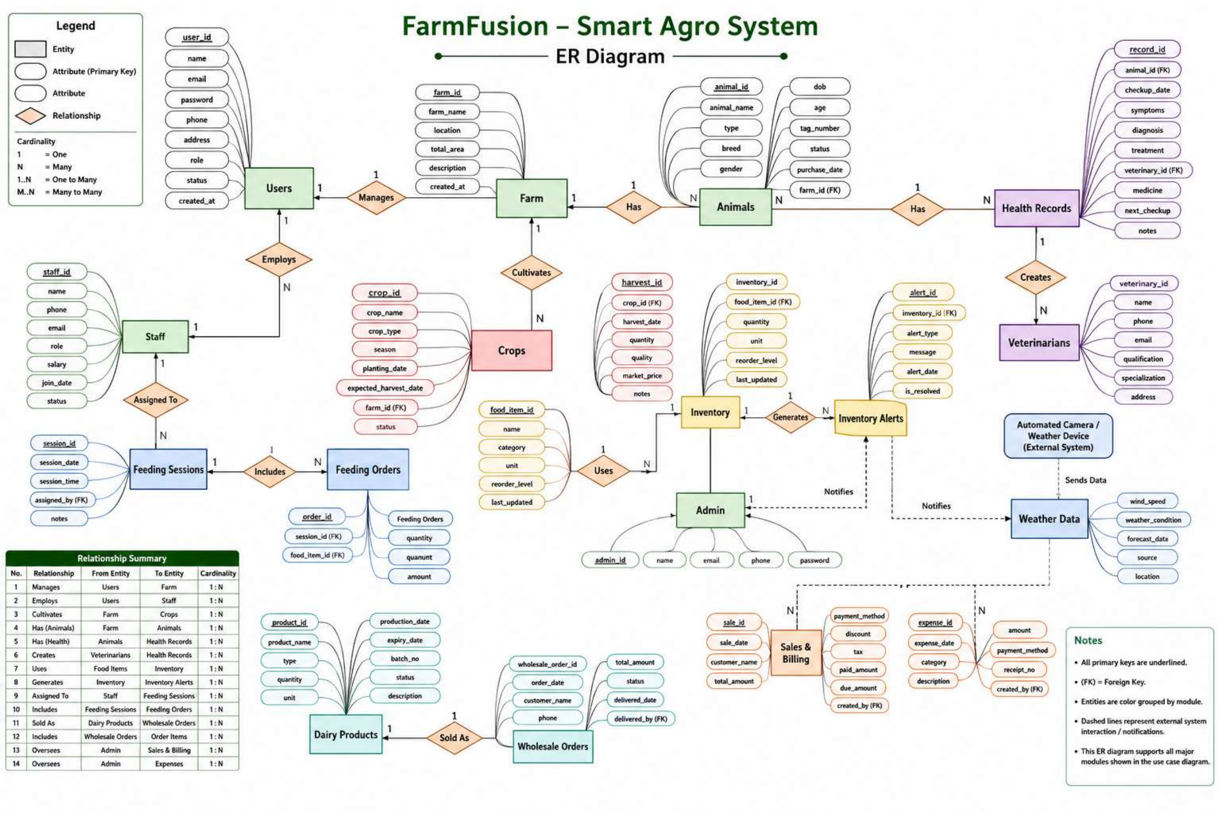

Figure 3.1.3: Entity-Relationship Diagram of the FarmFusion – Smart Agro System

The ERD models the persistent information needed by FarmFusion. It connects user roles to operational records and represents relationships among animals, health records, dairy production, inventory, crops, harvest records, alerts, orders, invoices, expenses and payments.

Reduction Table

Table Name	Primary Key	Foreign Key	Attributes
User	User ID	—	Name, Email, Password, Role, Phone, Farm Location
Farmer	Farmer ID	User ID	Farm Location, Contact Information
Veterinarian	Veterinarian ID	User ID	License Number
Livestock	Livestock ID	Farmer ID	Tag Number, Breed, Birth Date, Health Status
HealthRecord	Health Record ID	Livestock ID	Check Date, Diagnosis, Treatment
MilkProduction	Production ID	Livestock ID	Record Date, Quantity Liters
InventoryItem	Inventory ID	—	Item Name, Category, Quantity, Reorder Level
Alert	Alert ID	—	Type, Message, Timestamp, Recipient Role
Crop	Crop ID	Farmer ID	Crop Name, Planting Date, Expected Harvest
HarvestRecord	Harvest ID	Crop ID	Harvest Date, Yield Weight
DairyProduct	Dairy Product ID	—	Product Name, Price, Stock Quantity
SalesOrder	Order ID	Dairy Product ID	Order Date, Total Amount, Status
Invoice	Invoice ID	Order ID	Invoice Number, Amount, Issue/Due Date
Payment	Payment ID	Invoice ID	Amount, Method, Payment Date

Table Name	**Primary Key**	**Foreign Key**	**Attributes**
Expense	Expense ID	Admin/User ID	Description, Amount, Category, Expense Date
WeatherData	Weather ID	—	Temperature, Humidity, Wind Speed, Rainfall, Timestamp, Forecast

3.2 Behavioral & Interaction Design

3.2.1 Activity Diagram

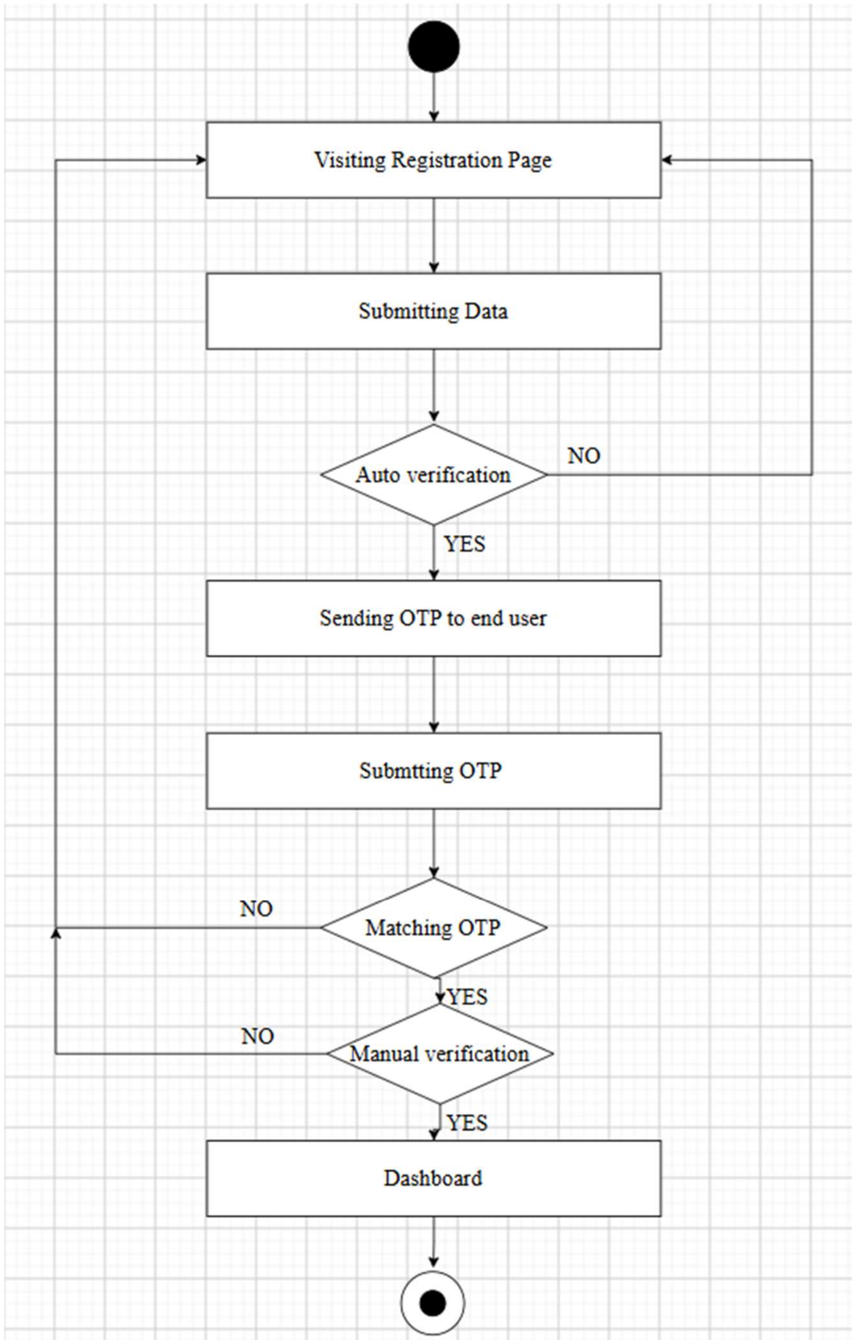

Figure 3.2.1: Activity Diagram for User Registration

The activity begins at the registration page, continues through data submission and verification, sends an OTP, verifies the OTP and finally opens the dashboard after successful account creation.

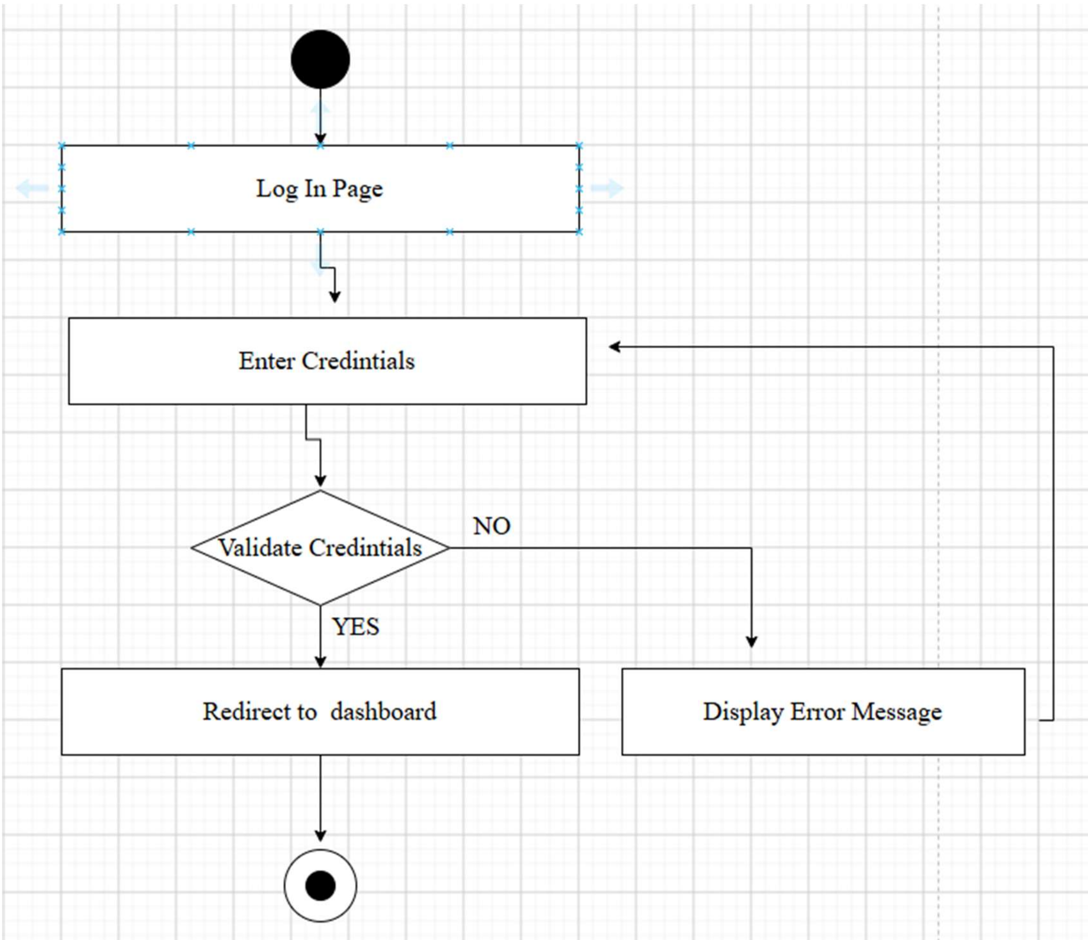

Figure 3.2.2: Activity Diagram for User Login

The user enters credentials, the system validates them and either redirects to the dashboard or returns an error for another attempt.

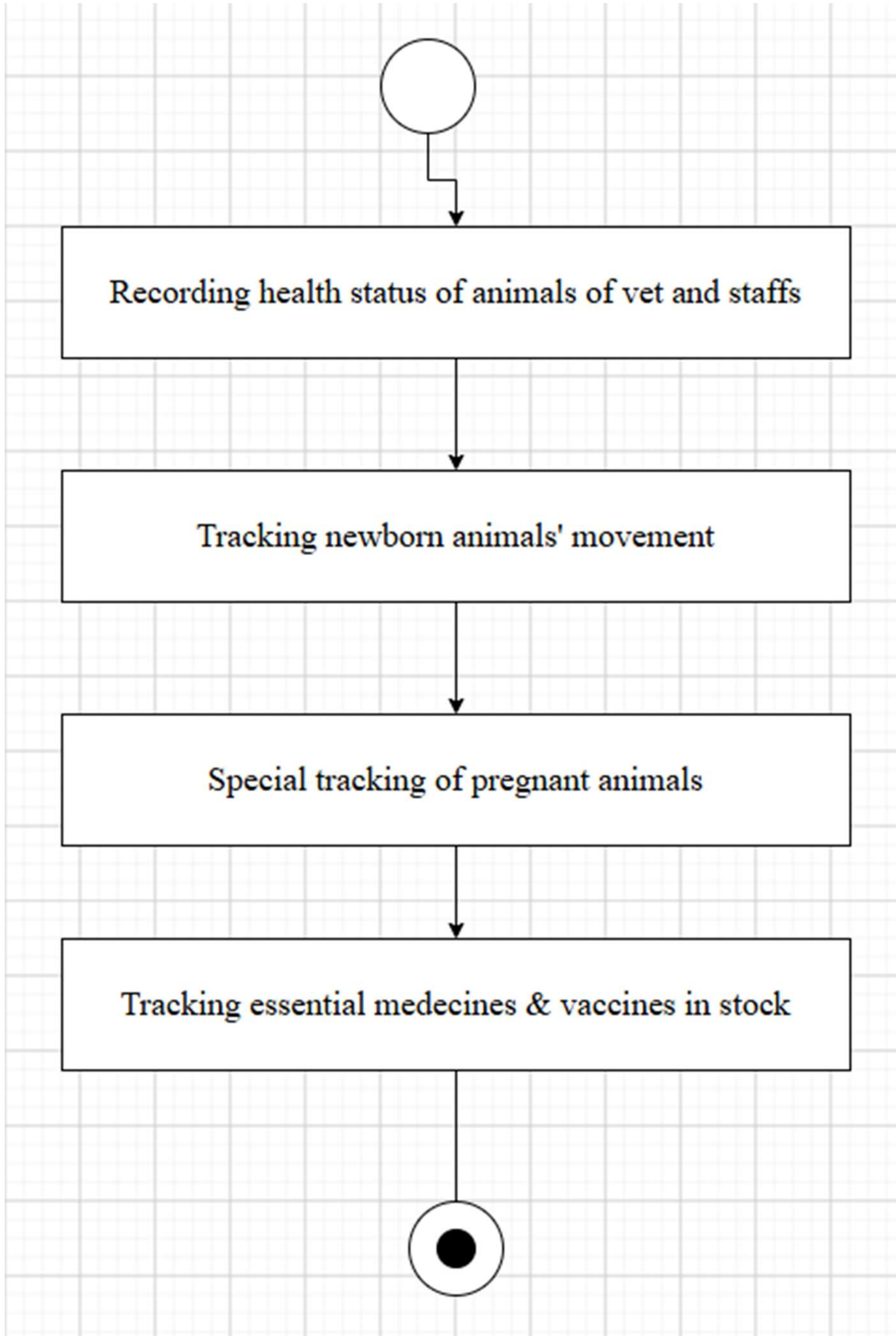

Figure 3.2.3: Activity Diagram for Animal Health Monitoring

The workflow records animal health, tracks newborn movement and pregnancy status, and checks essential medicine and vaccine stock.

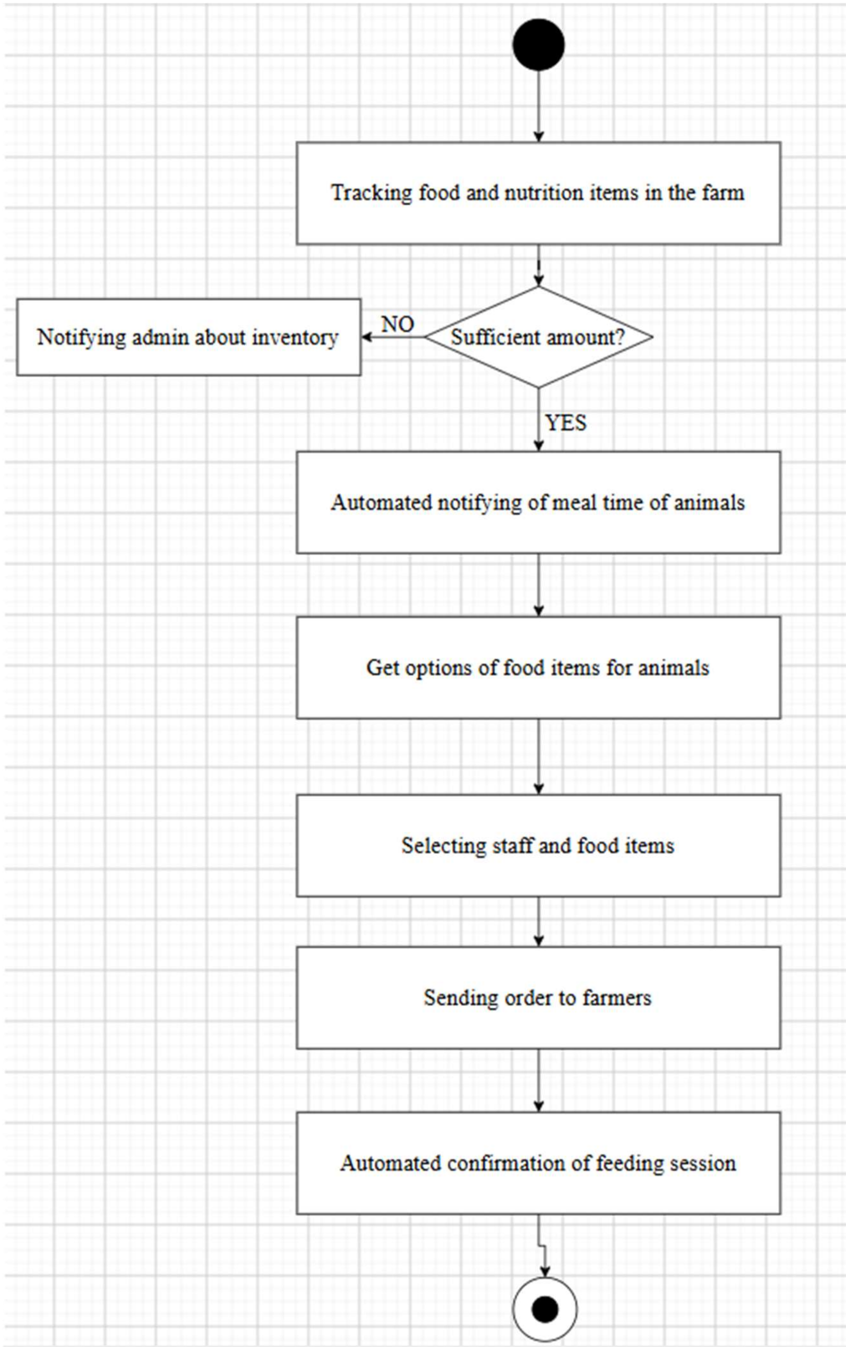

Figure 3.2.4: Activity Diagram for Automated Inventory / Feeding Workflow

The workflow checks whether sufficient feed is available, generates an inventory alert when necessary, schedules meal notification, selects feed and staff, sends the feeding order and confirms the session.

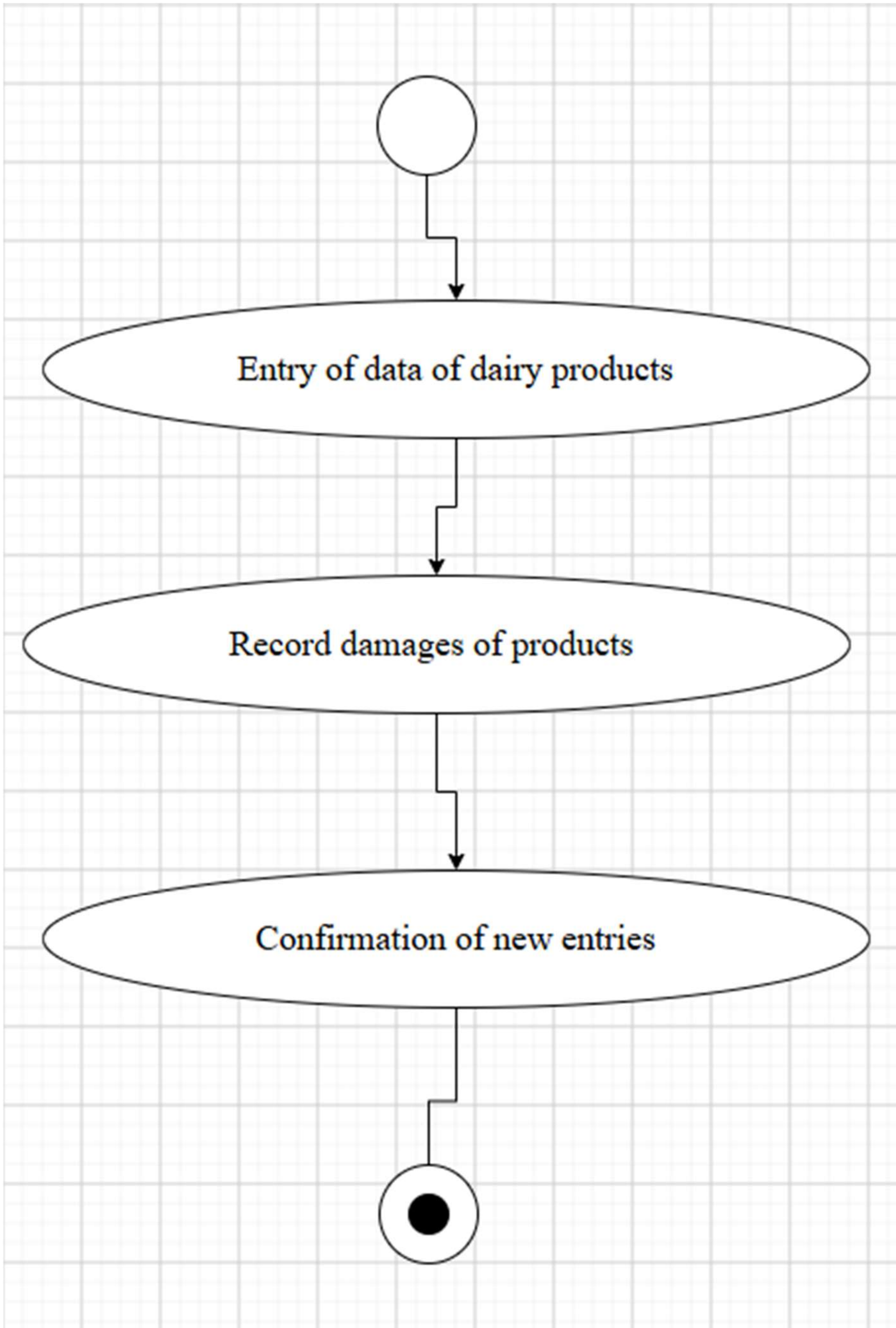

Figure 3.2.5: Activity Diagram for Dairy Product Management

The workflow records dairy-product details, records damage or quantity information and confirms the new entry.

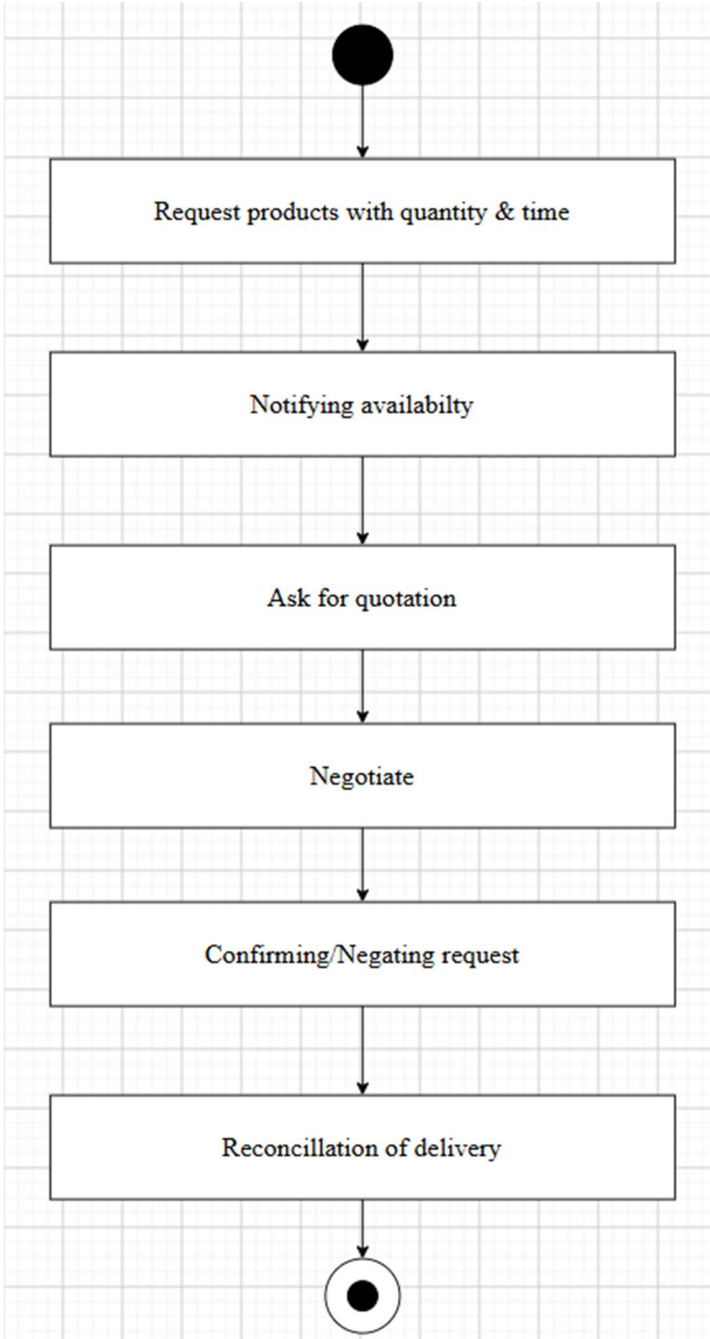

Figure 3.2.6: Activity Diagram for Dairy Product Wholesale Management

The workflow receives a product request, checks availability, asks for quotation, negotiates, confirms or rejects the request and reconciles delivery.

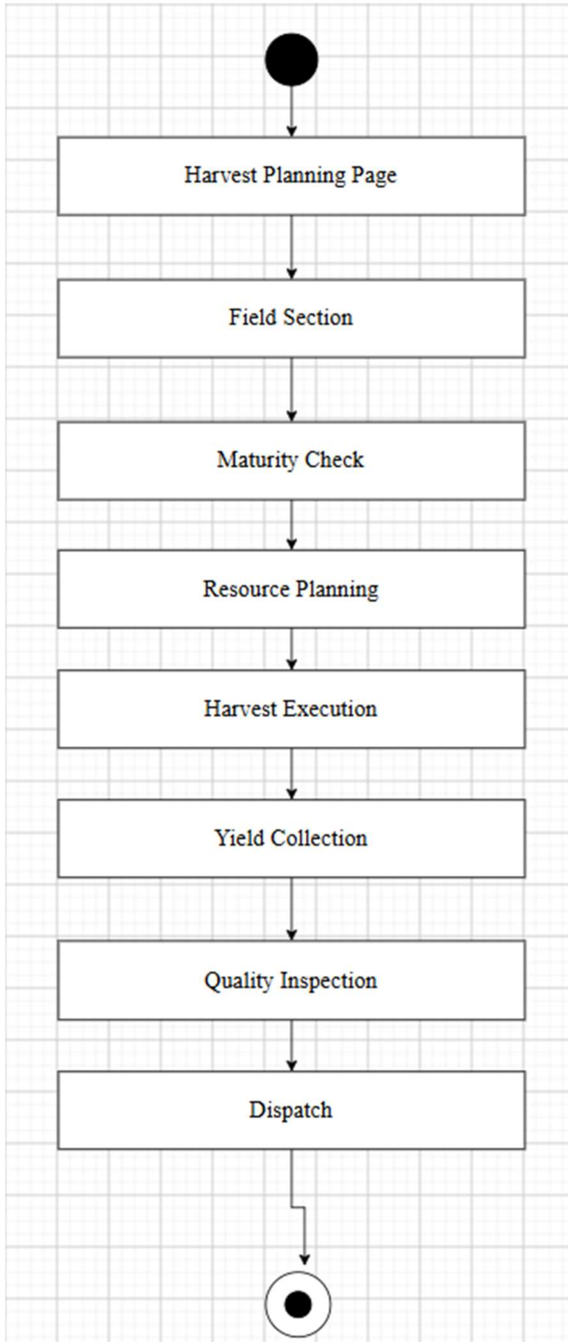

Figure 3.2.7: Activity Diagram for Crop Harvesting Management

The workflow moves from harvest planning to field selection, maturity check, resource planning, execution, yield collection, quality inspection and dispatch.

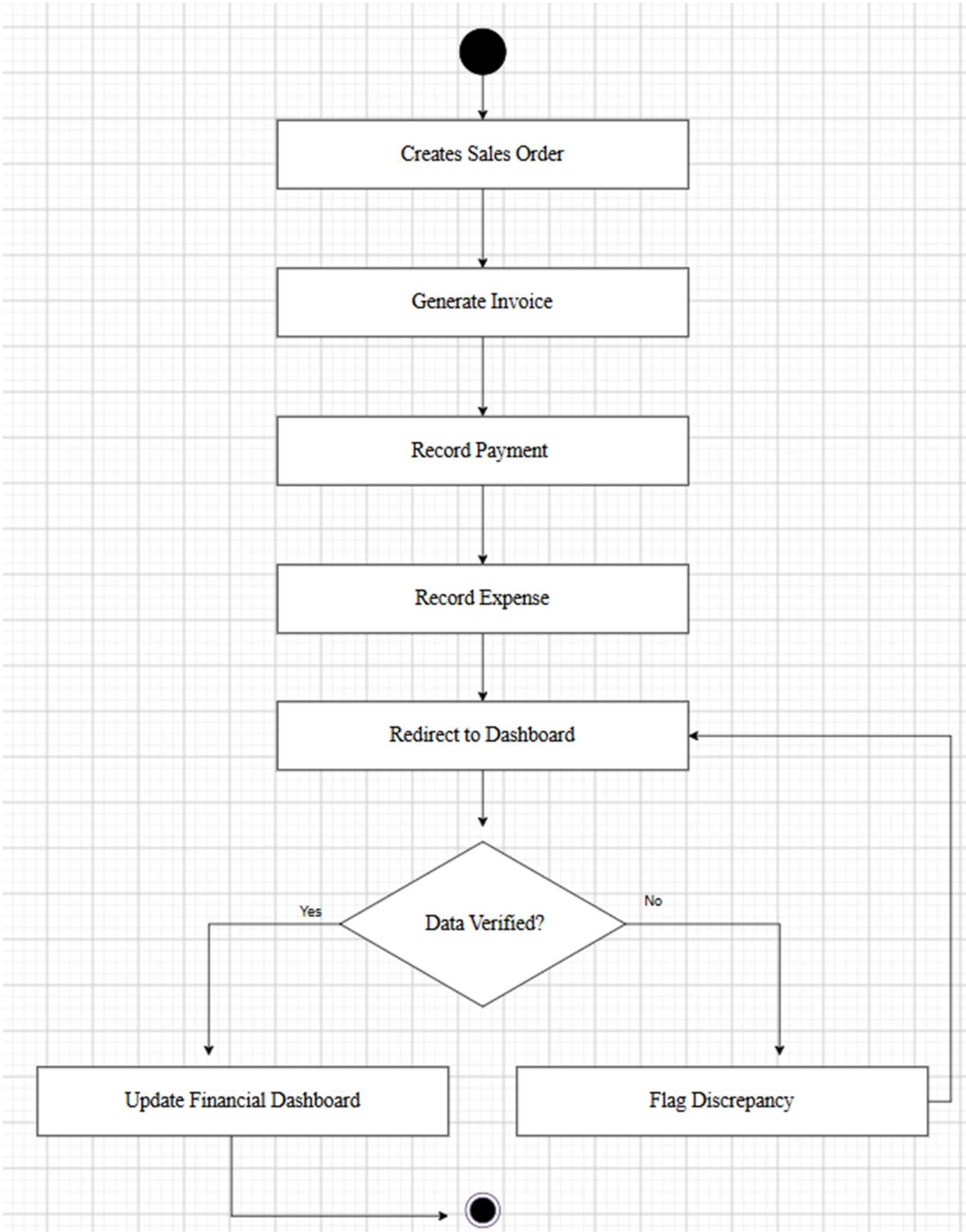

Figure 3.2.8: Activity Diagram for Sales, Billing & Expense Tracking

The workflow creates a sales order, generates an invoice, records payment and expense information, verifies the financial data and either updates the dashboard or flags a discrepancy.

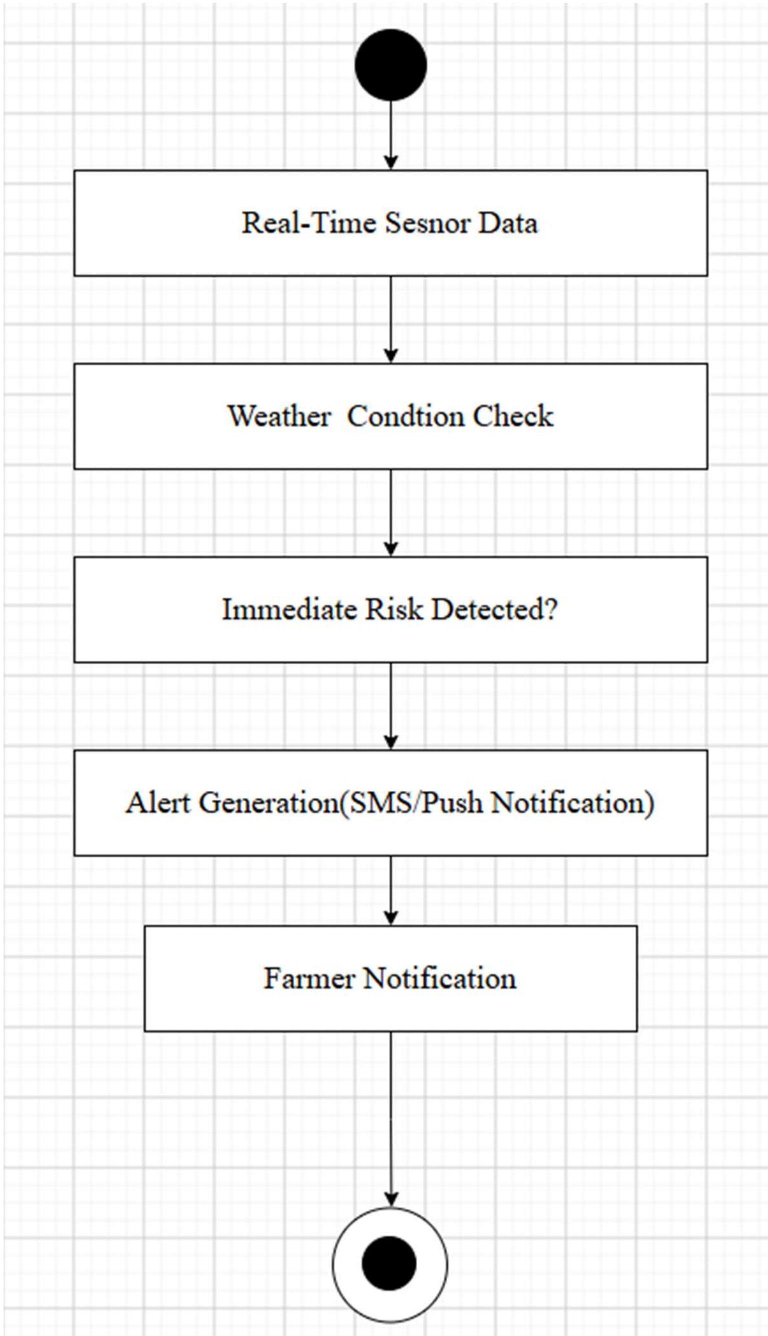

Figure 3.2.9: Activity Diagram for Weather Monitoring & Alert

The workflow receives real-time sensor data, checks conditions, detects immediate risk, generates an alert and notifies the farmer.

3.2.2 Sequence Diagram

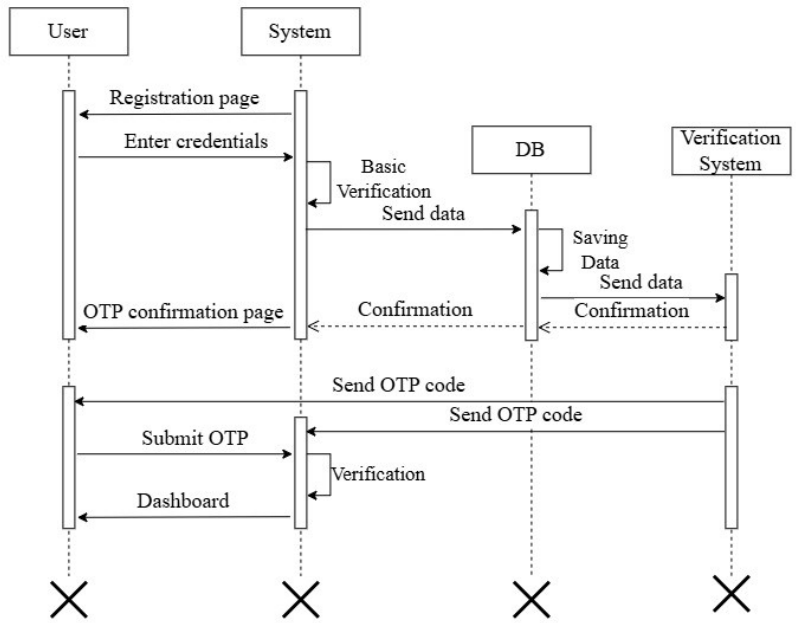

Figure 3.2.10: Sequence Diagram for User Registration

The sequence diagram for User Registration shows the ordered interaction among the relevant actor, FarmFusion system services, database and automated/external components. It is consistent with the project's use case and activity models and demonstrates how requests, responses, updates and notifications are exchanged.

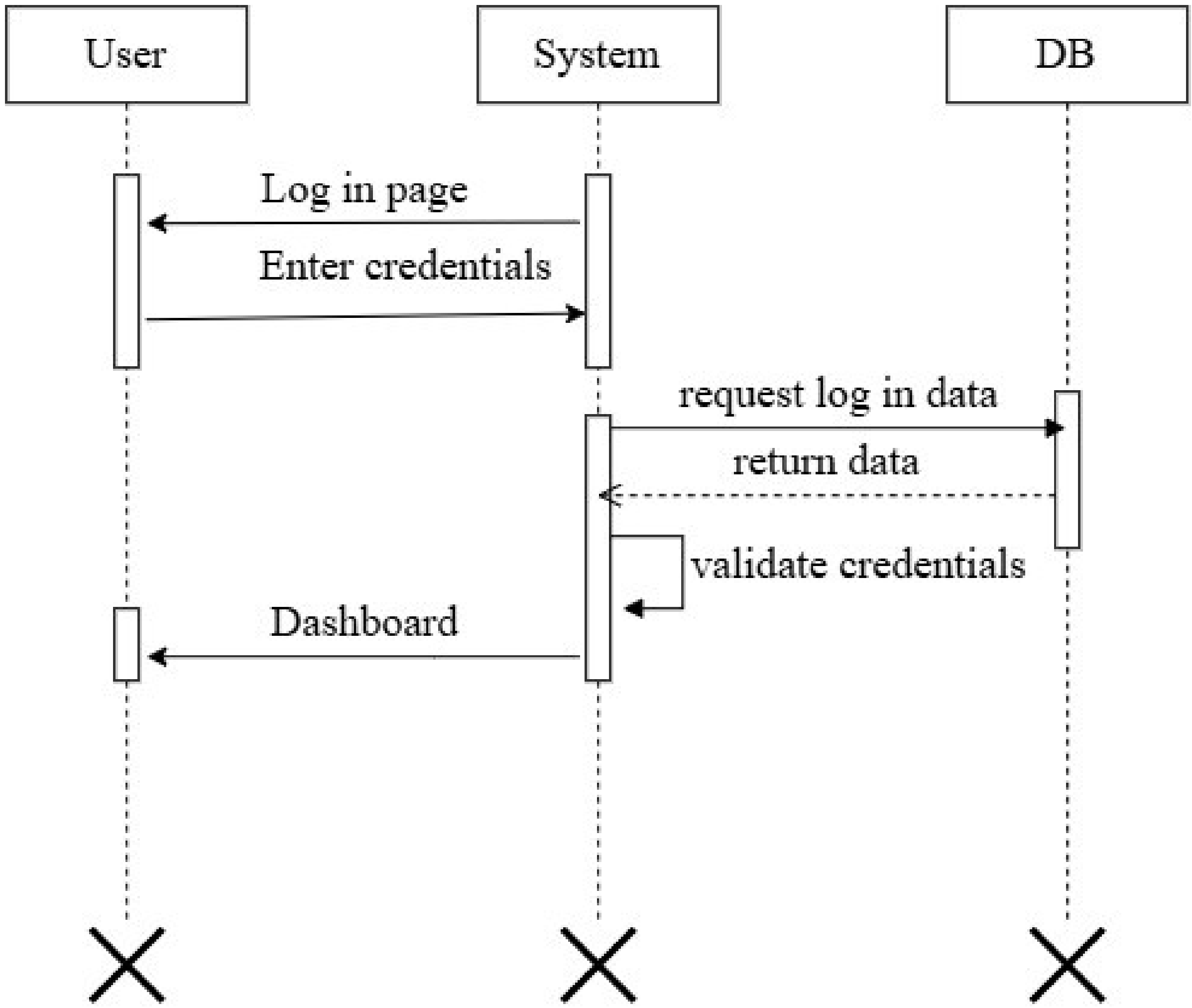

Figure 3.2.11: Sequence Diagram for User Login

The sequence diagram for User Login shows the ordered interaction among the relevant actor, FarmFusion system services, database and automated/external components. It is consistent with the project's use case and activity models and demonstrates how requests, responses, updates and notifications are exchanged.

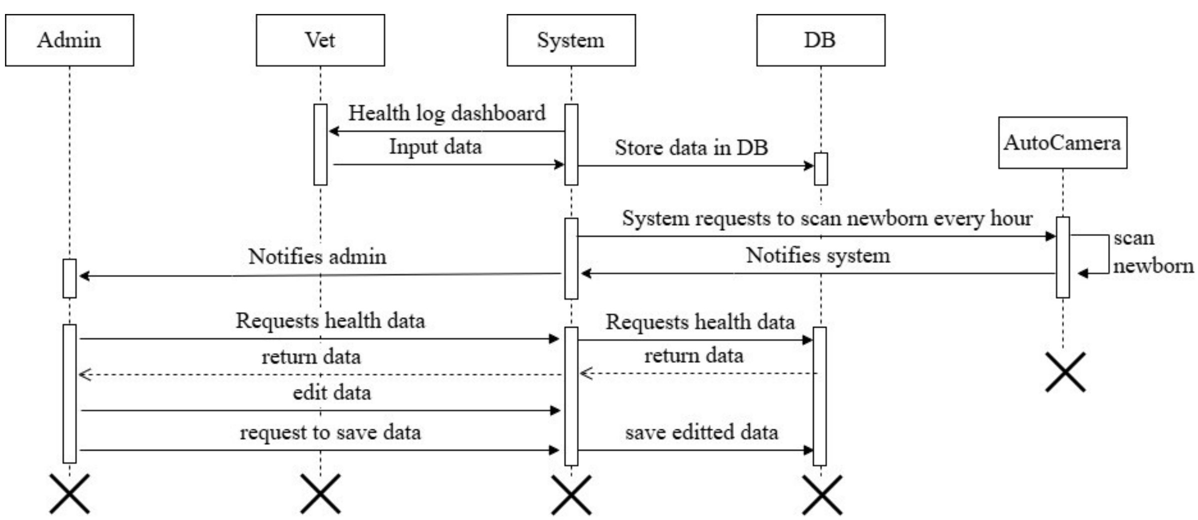

Figure 3.2.12: Sequence Diagram for Animal Health Monitoring

The sequence diagram for Animal Health Monitoring shows the ordered interaction among the relevant actor, FarmFusion system services, database and automated/external components. It is consistent with the project's use case and activity models and demonstrates how requests, responses, updates and notifications are exchanged.

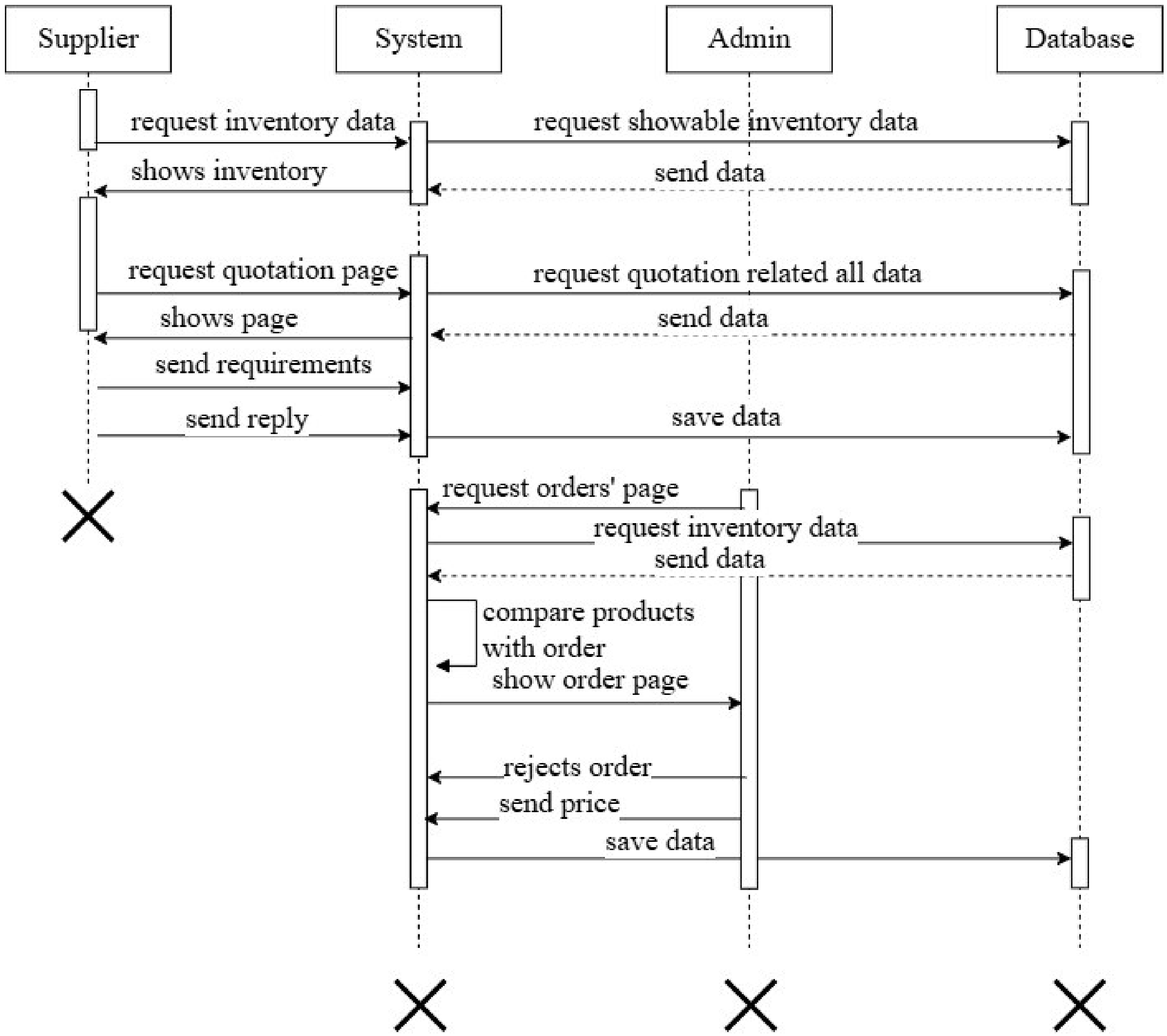

Figure 3.2.13: Sequence Diagram for Automated Inventory Management

The sequence diagram for Automated Inventory Management shows the ordered interaction among the relevant actor, FarmFusion system services, database and automated/external components. It is consistent with the project's use case and activity models and demonstrates how requests, responses, updates and notifications are exchanged.

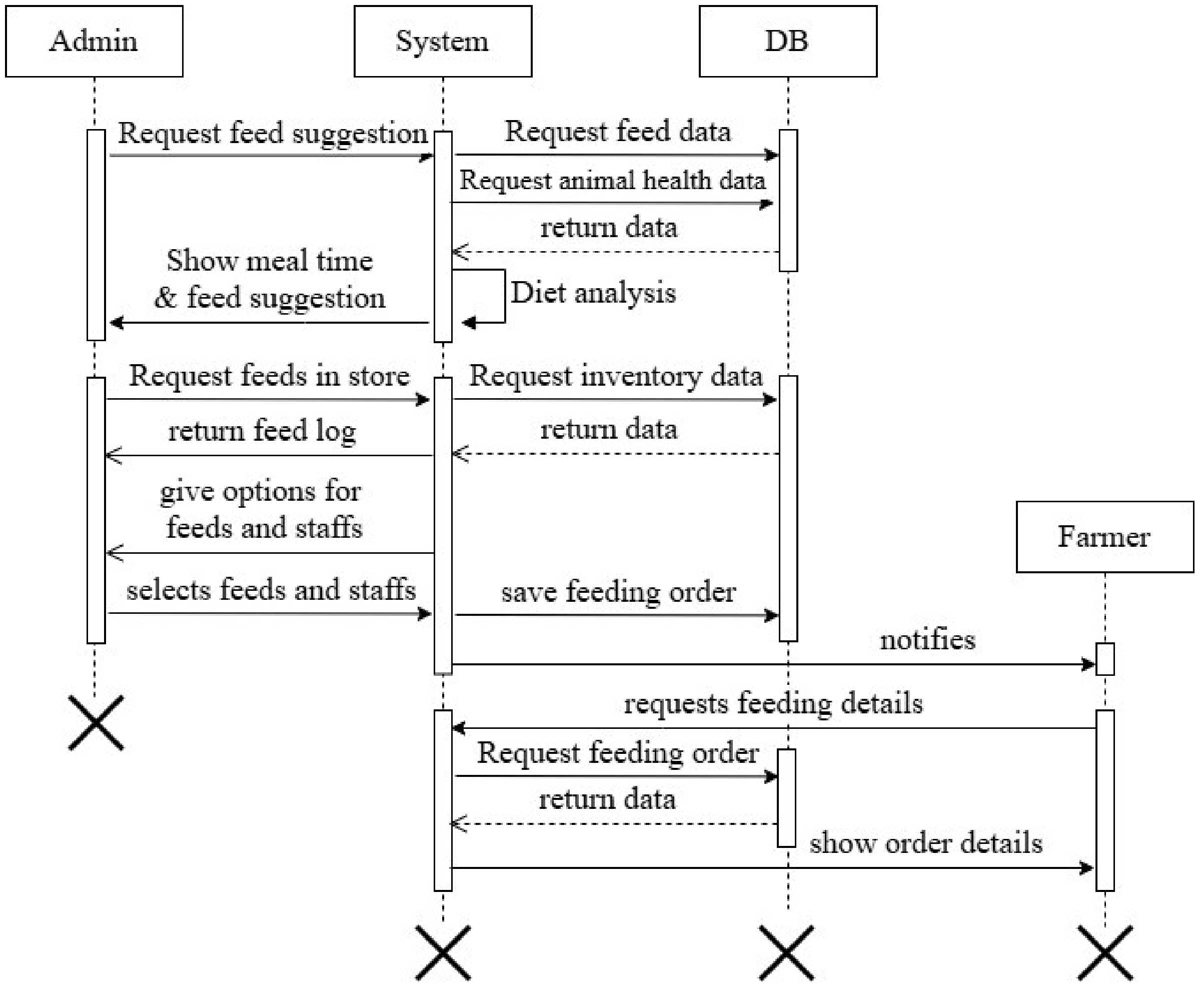

Figure 3.2.14: Sequence Diagram for Feeding & Nutrition Management

The sequence diagram for Feeding & Nutrition Management shows the ordered interaction among the relevant actor, FarmFusion system services, database and automated/external components. It is consistent with the project's use case and activity models and demonstrates how requests, responses, updates and notifications are exchanged.

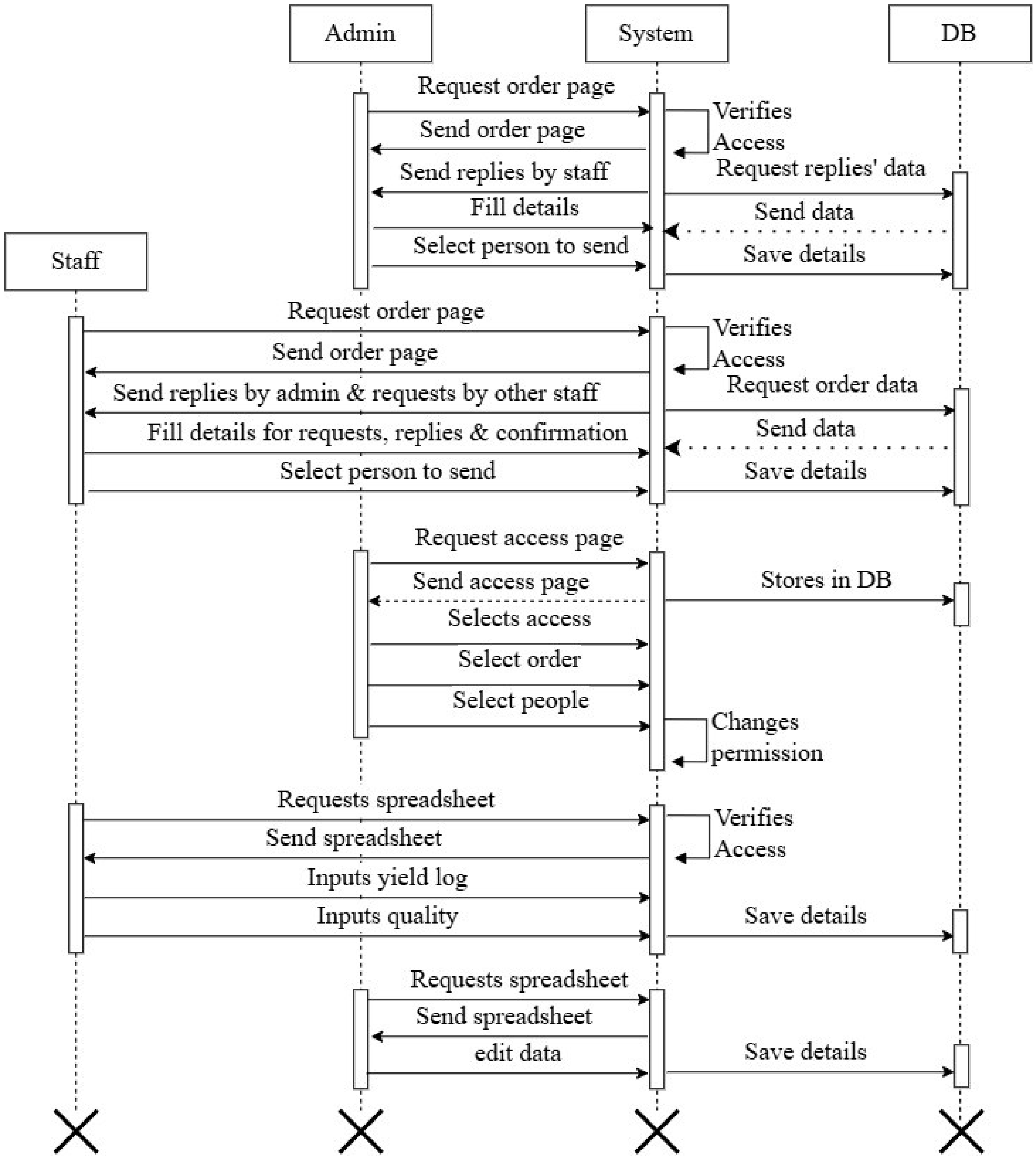

Figure 3.2.15: Sequence Diagram for Crop Harvesting Management

The sequence diagram for Crop Harvesting Management shows the ordered interaction among the relevant actor, FarmFusion system services, database and automated/external components. It is consistent with the project's use case and activity models and demonstrates how requests, responses, updates and notifications are exchanged.

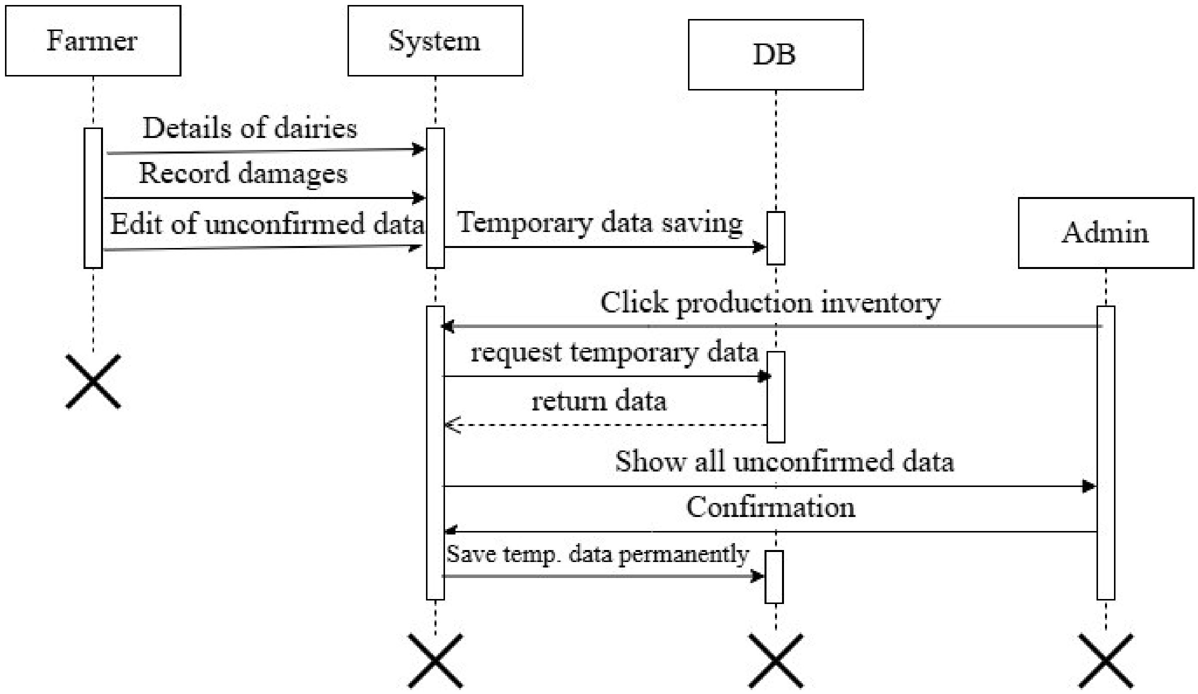

Figure 3.2.16: Sequence Diagram for Dairy Product Management

The sequence diagram for Dairy Product Management shows the ordered interaction among the relevant actor, FarmFusion system services, database and automated/external components. It is consistent with the project's use case and activity models and demonstrates how requests, responses, updates and notifications are exchanged.

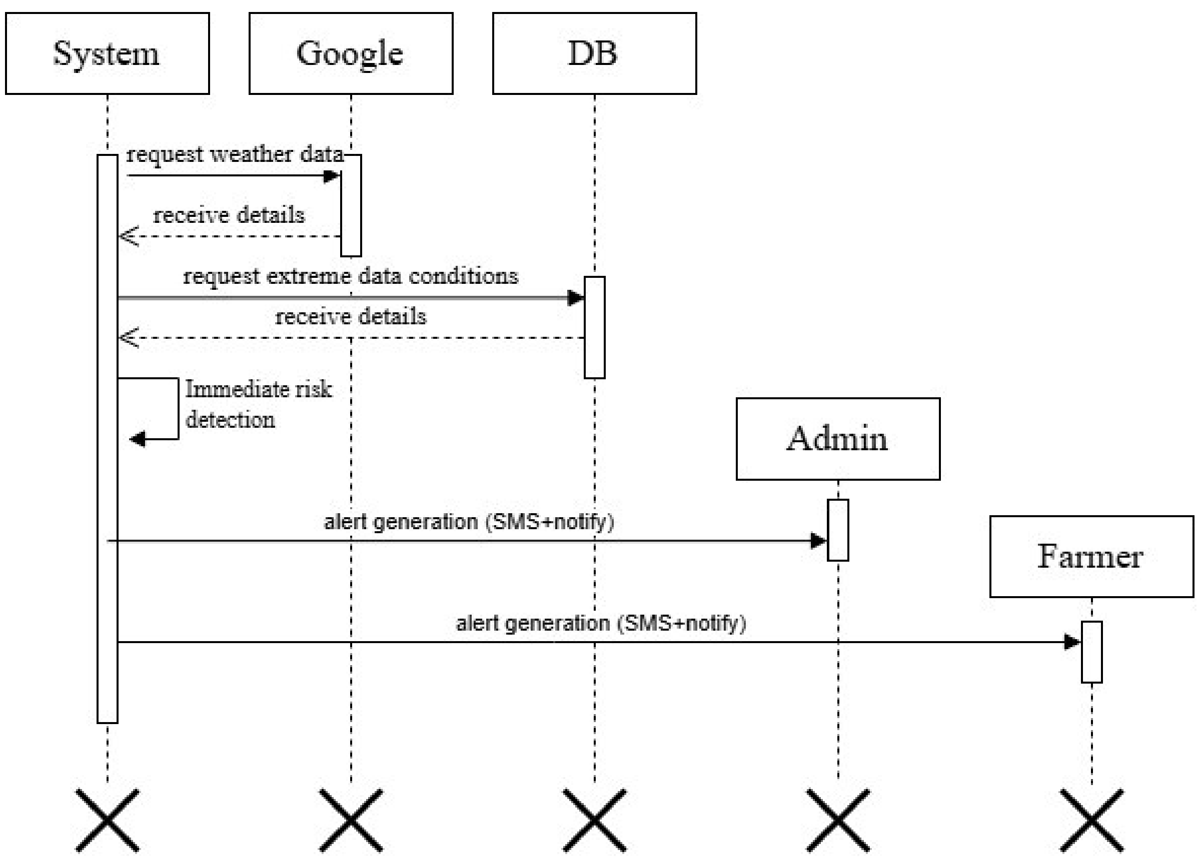

Figure 3.2.17: Sequence Diagram for Weather Monitoring & Alert

The sequence diagram for Weather Monitoring & Alert shows the ordered interaction among the relevant actor, FarmFusion system services, database and automated/external components. It is consistent with the project's use case and activity models and demonstrates how requests, responses, updates and notifications are exchanged.

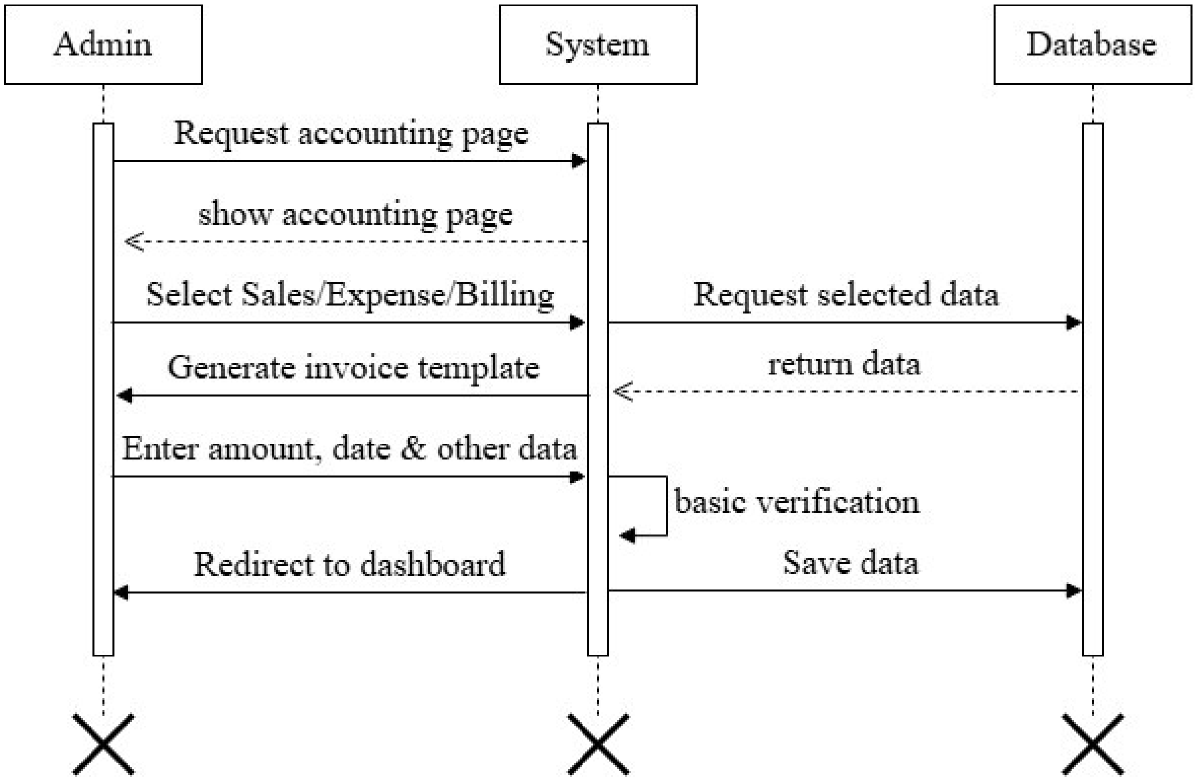

Figure 3.2.18: Sequence Diagram for Sales, Billing & Expense Tracking

The sequence diagram for Sales, Billing & Expense Tracking shows the ordered interaction among the relevant actor, FarmFusion system services, database and automated/external components. It is consistent with the project's use case and activity models and demonstrates how requests, responses, updates and notifications are exchanged.

3.2.3 State Diagram

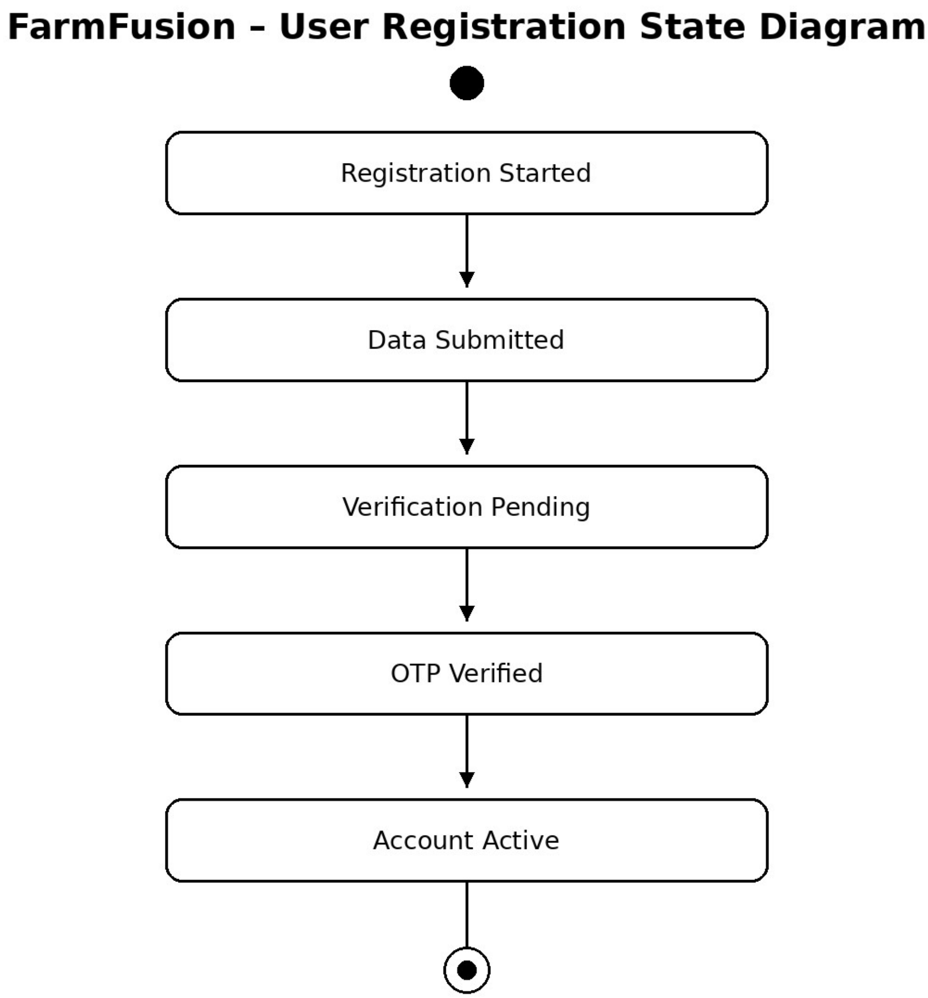

Figure 3.3.1: State Diagram for User Registration

The state model for User Registration describes the major lifecycle stages of the process, from initiation through validation or processing to the final operational state. The transitions correspond to the workflow represented in the project's activity and sequence diagrams.

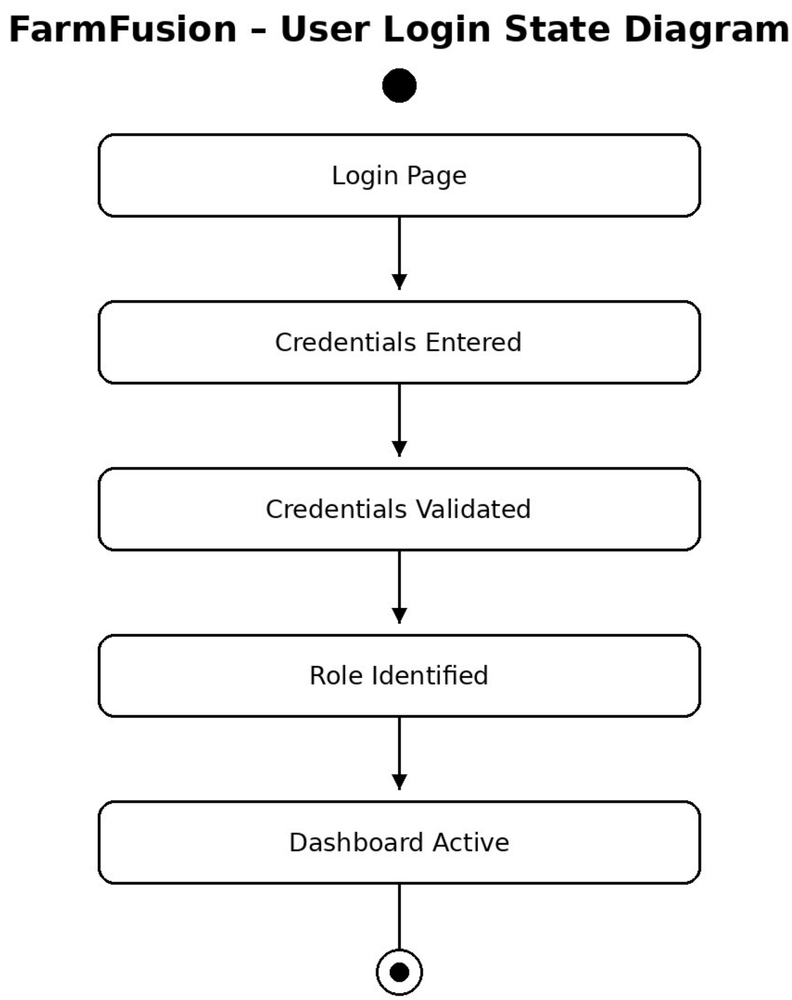

Figure 3.3.2: State Diagram for User Login

The state model for User Login describes the major lifecycle stages of the process, from initiation through validation or processing to the final operational state. The transitions correspond to the workflow represented in the project's activity and sequence diagrams.

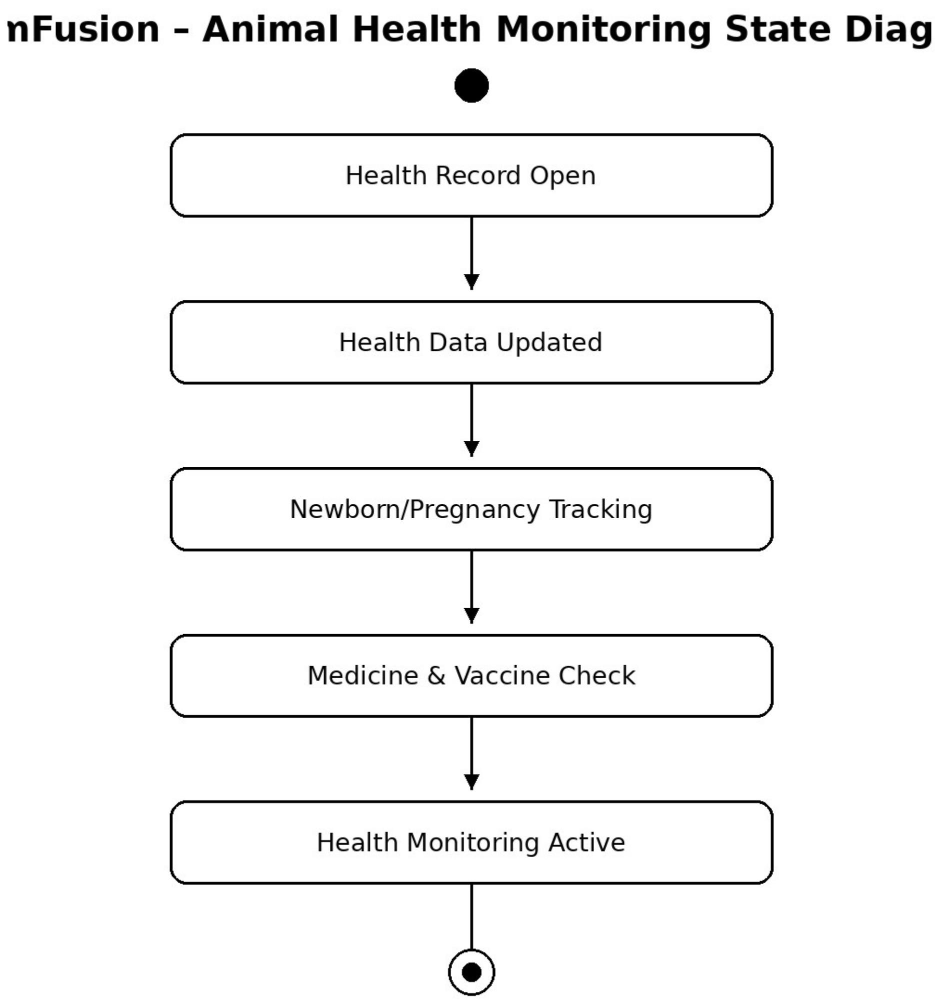

Figure 3.3.3: State Diagram for Animal Health Monitoring

The state model for Animal Health Monitoring describes the major lifecycle stages of the process, from initiation through validation or processing to the final operational state. The transitions correspond to the workflow represented in the project's activity and sequence diagrams.

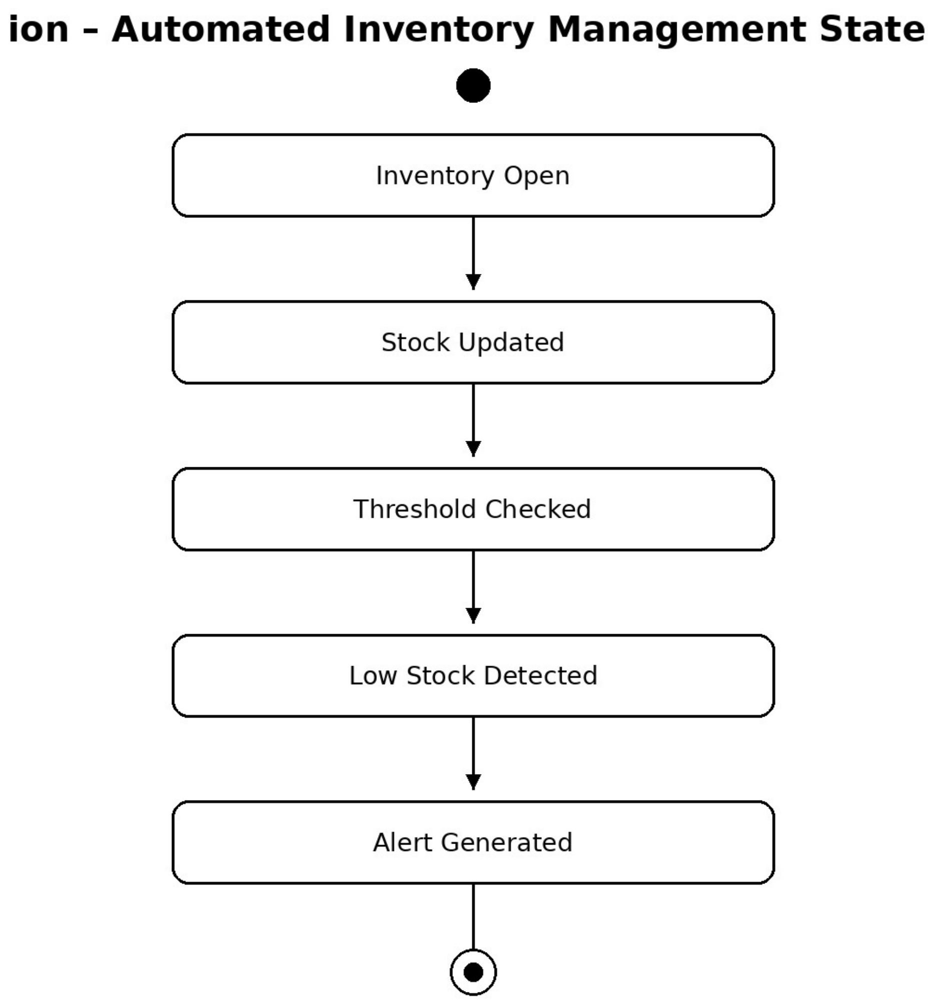

Figure 3.3.4: State Diagram for Automated Inventory Management

The state model for Automated Inventory Management describes the major lifecycle stages of the process, from initiation through validation or processing to the final operational state. The transitions correspond to the workflow represented in the project's activity and sequence diagrams.

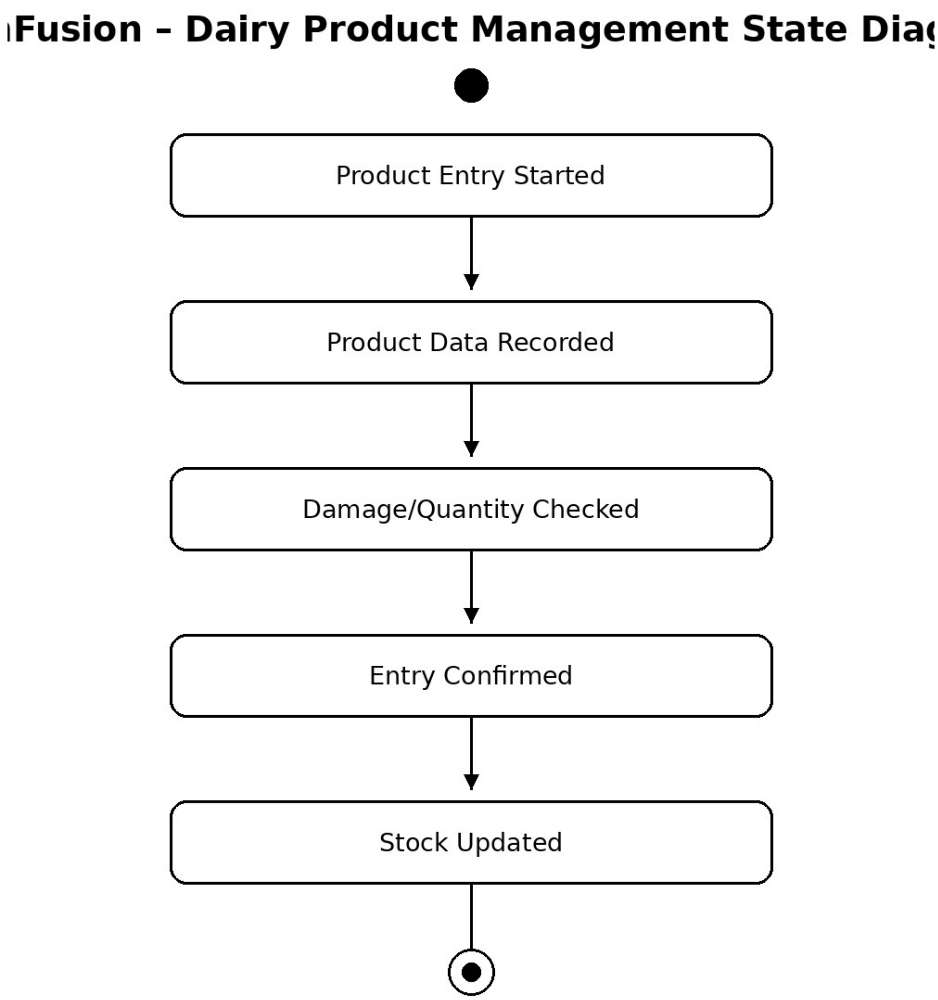

Figure 3.3.5: State Diagram for Dairy Product Management

The state model for Dairy Product Management describes the major lifecycle stages of the process, from initiation through validation or processing to the final operational state. The transitions correspond to the workflow represented in the project's activity and sequence diagrams.

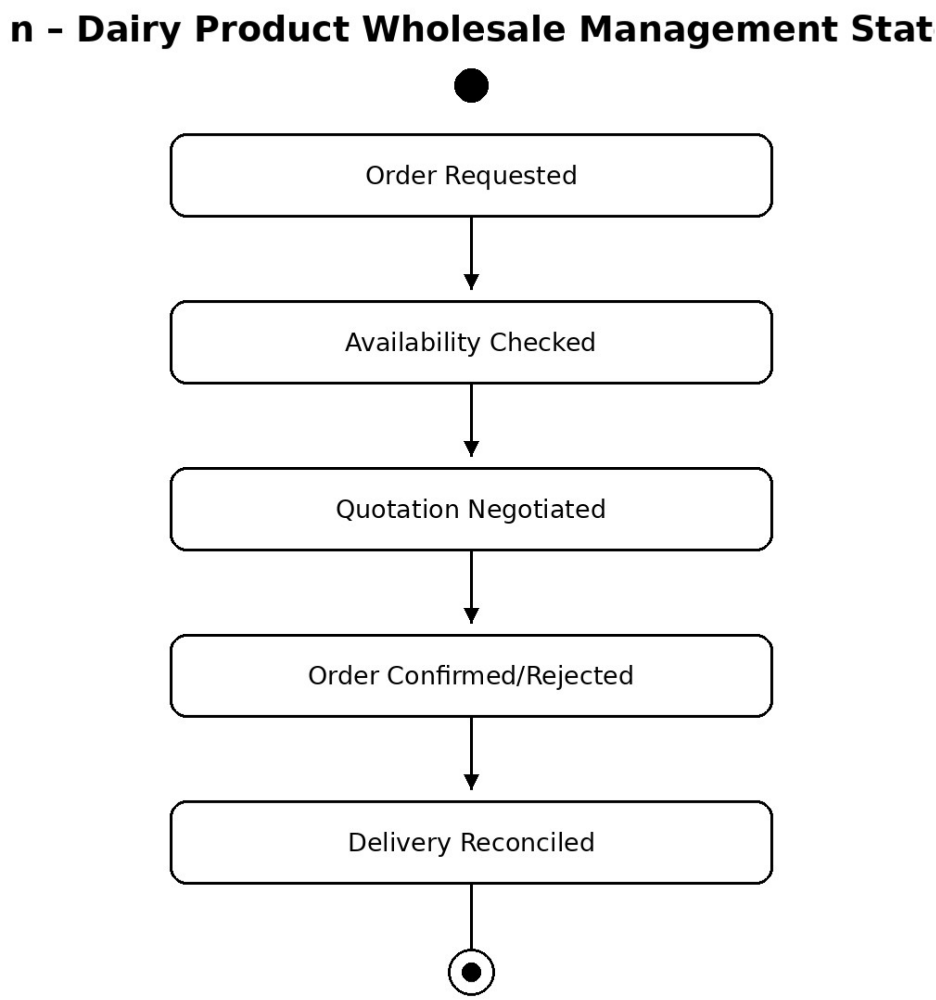

Figure 3.3.6: State Diagram for Dairy Product Wholesale Management

The state model for Dairy Product Wholesale Management describes the major lifecycle stages of the process, from initiation through validation or processing to the final operational state. The transitions correspond to the workflow represented in the project's activity and sequence diagrams.

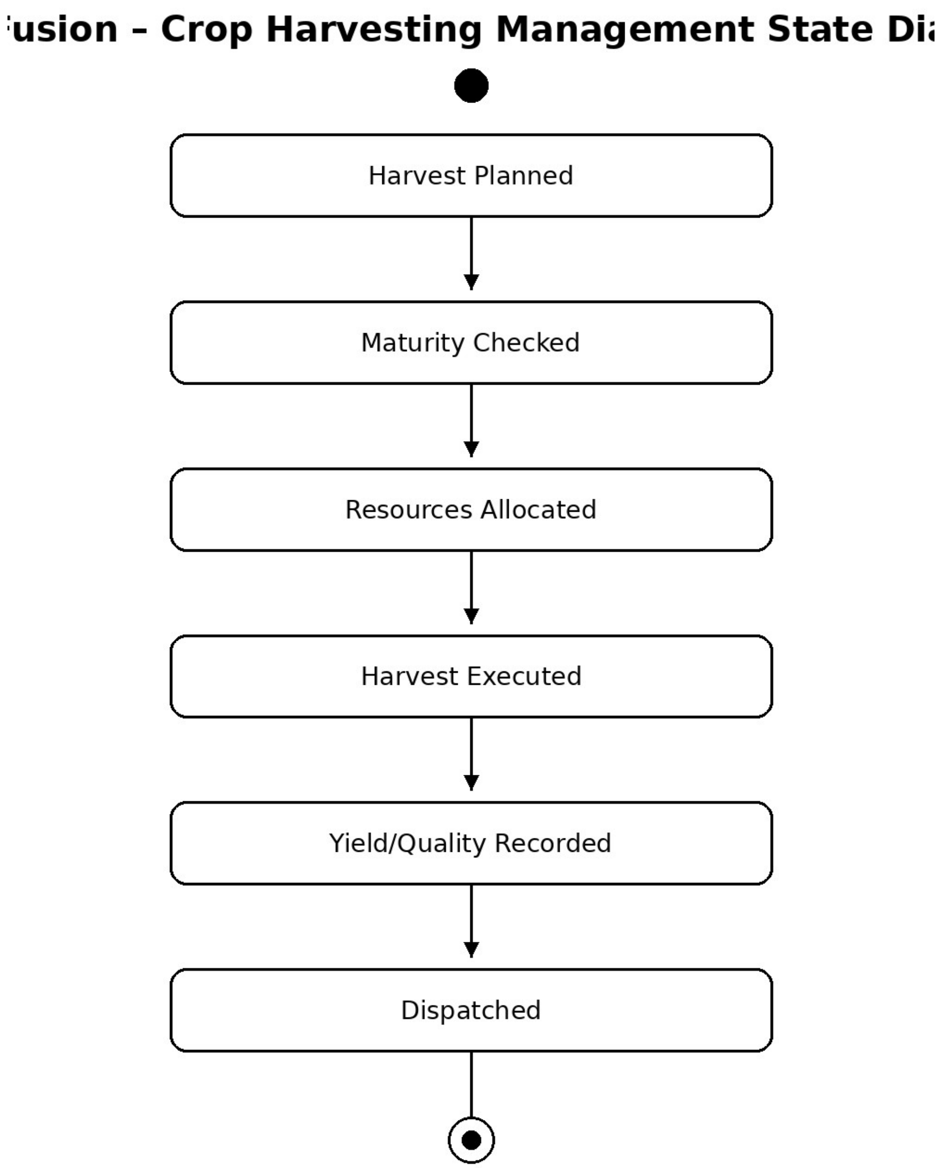

Figure 3.3.7: State Diagram for Crop Harvesting Management

The state model for Crop Harvesting Management describes the major lifecycle stages of the process, from initiation through validation or processing to the final operational state. The transitions correspond to the workflow represented in the project's activity and sequence diagrams.

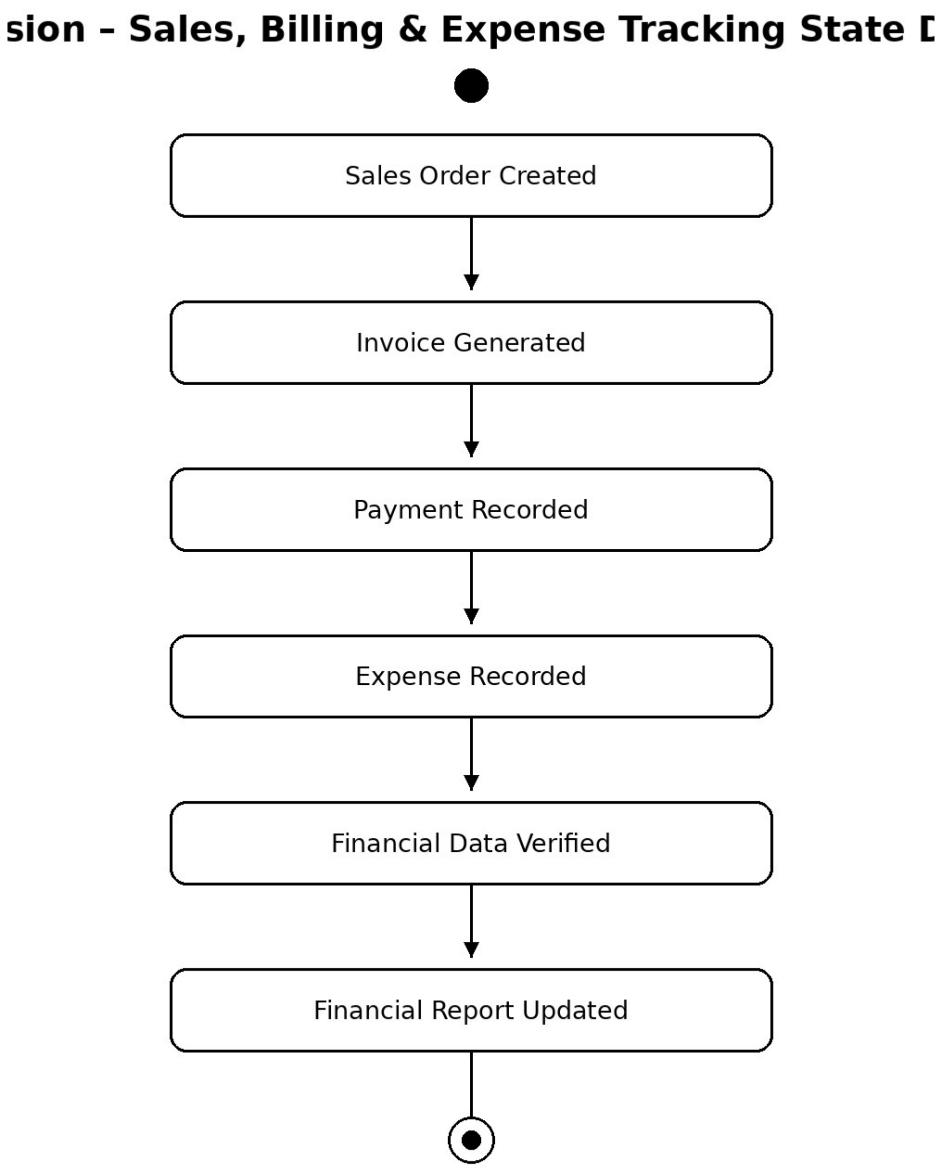

Figure 3.3.8: State Diagram for Sales, Billing & Expense Tracking

The state model for Sales, Billing & Expense Tracking describes the major lifecycle stages of the process, from initiation through validation or processing to the final operational state. The transitions correspond to the workflow represented in the project's activity and sequence diagrams.

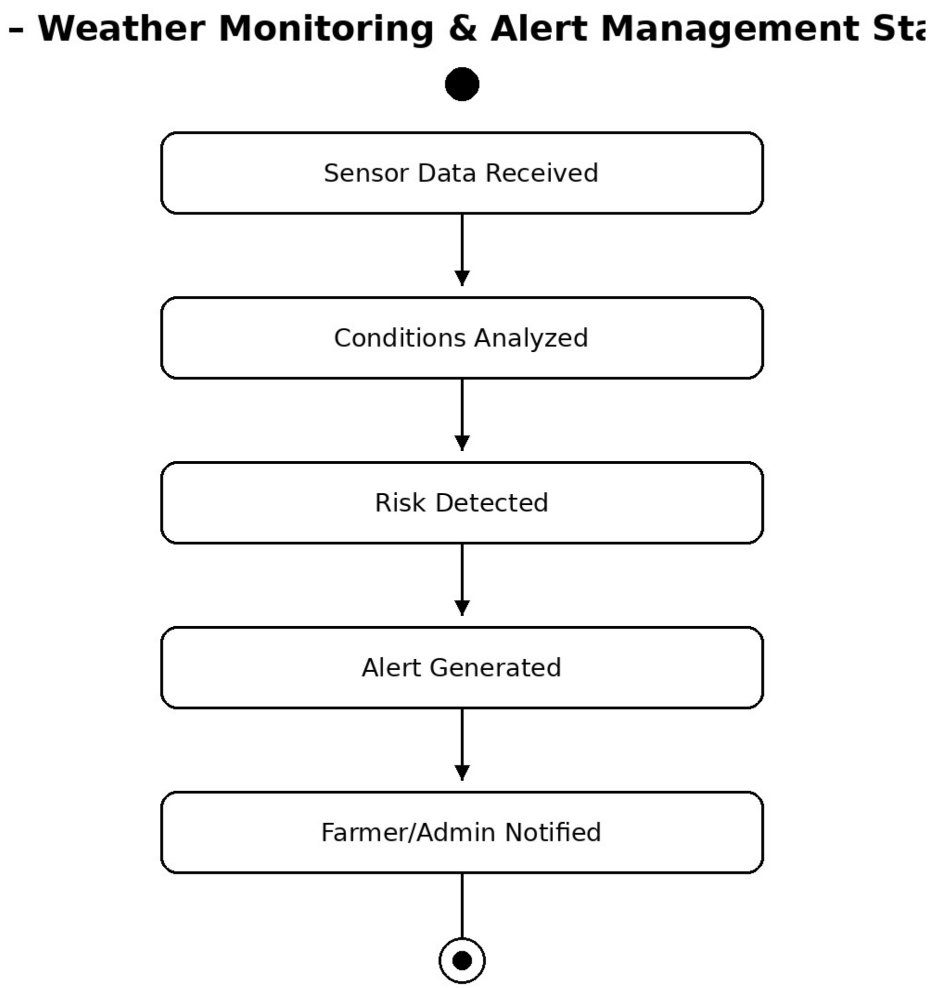

Figure 3.3.9: State Diagram for Weather Monitoring & Alert Management

The state model for Weather Monitoring & Alert Management describes the major lifecycle stages of the process, from initiation through validation or processing to the final operational state. The transitions correspond to the workflow represented in the project's activity and sequence diagrams.

3.3 Prototypes

The prototype section demonstrates how the system design can be translated into role-oriented interfaces. The prototype views below are conceptual interface mockups based on the FarmFusion modules and are intended to illustrate navigation, data cards and operational dashboards.

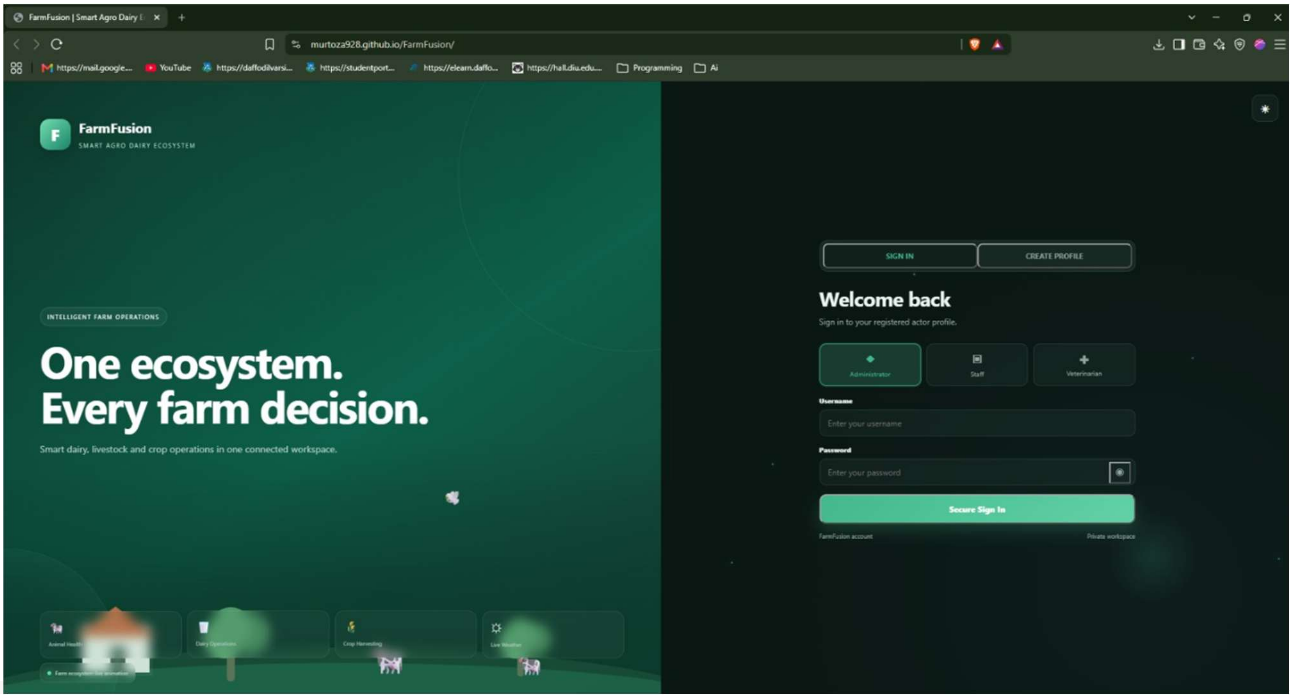

Figure 3.4.1: FarmFusion Overview Prototype

The FarmFusion Overview prototype emphasizes the information and actions most relevant to the selected role/module while keeping navigation consistent across the platform.

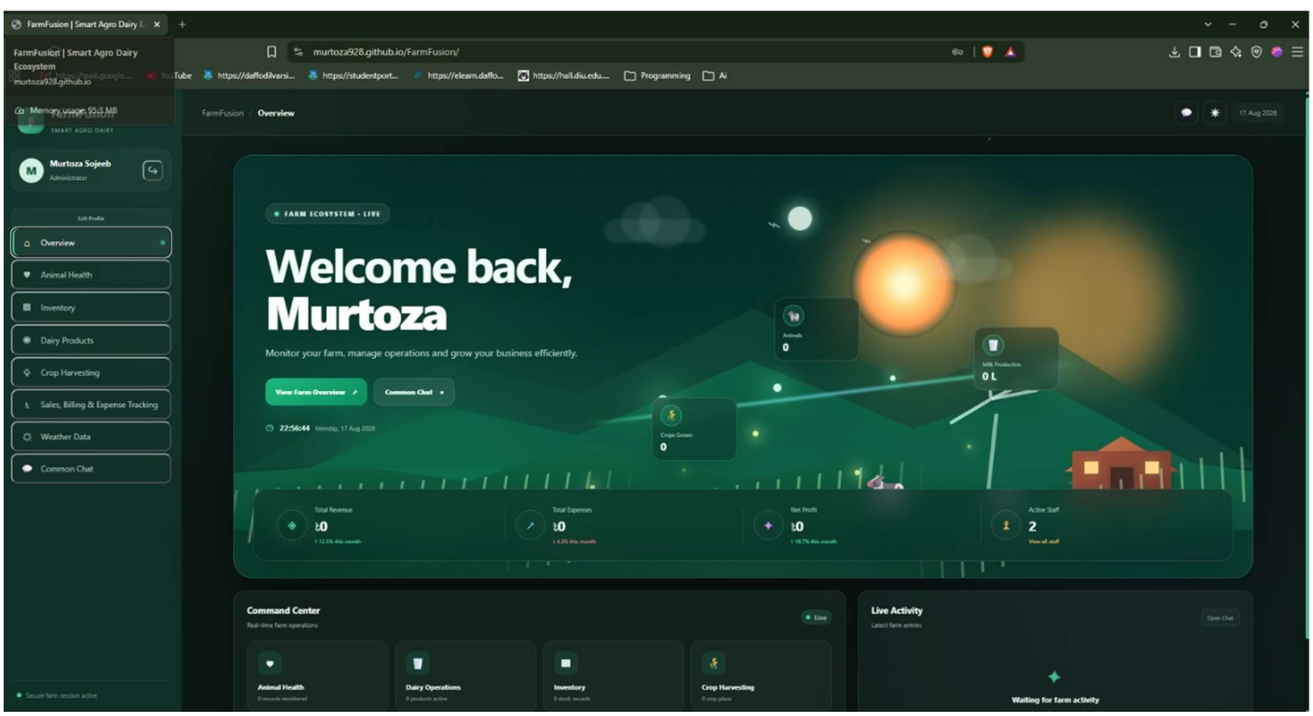

Figure 3.4.2: Admin Command Console Prototype

The Admin Command Console prototype emphasizes the information and actions most relevant to the selected role/module while keeping navigation consistent across the platform.

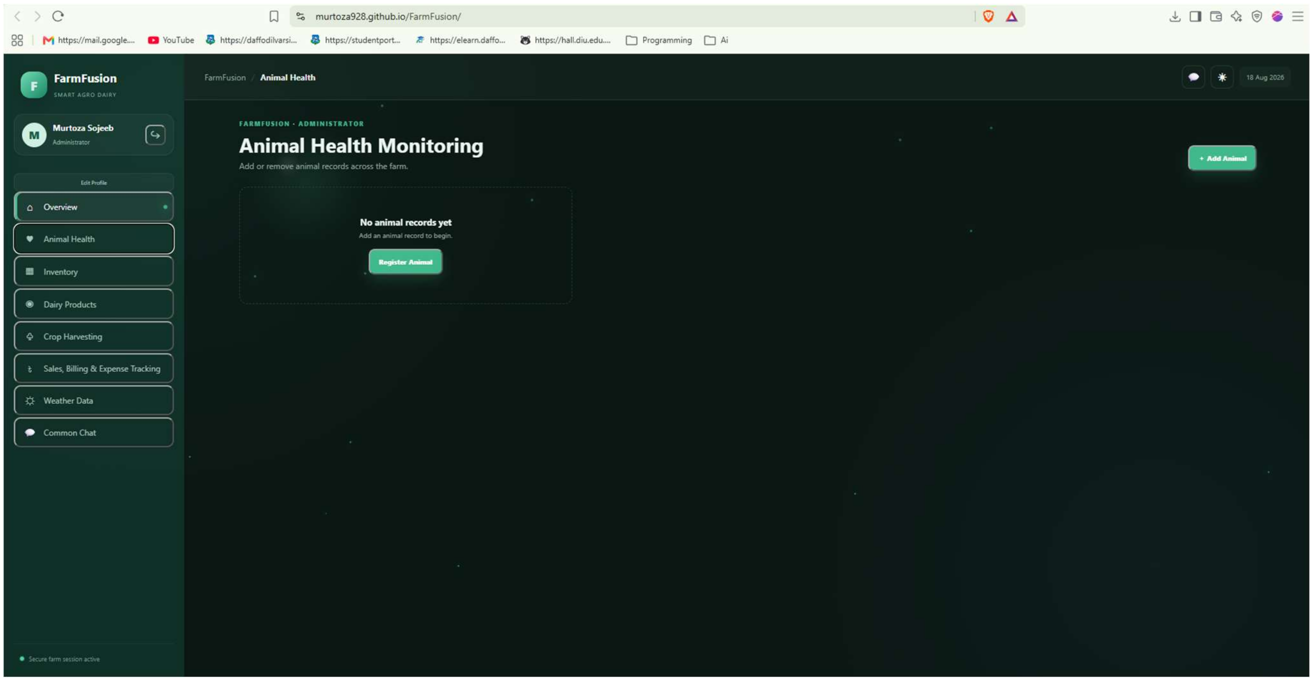

Figure 3.4.3: Animal Health Dashboard Prototype

The Animal Health Dashboard prototype emphasizes the information and actions most relevant to the selected role/module while keeping navigation consistent across the platform.

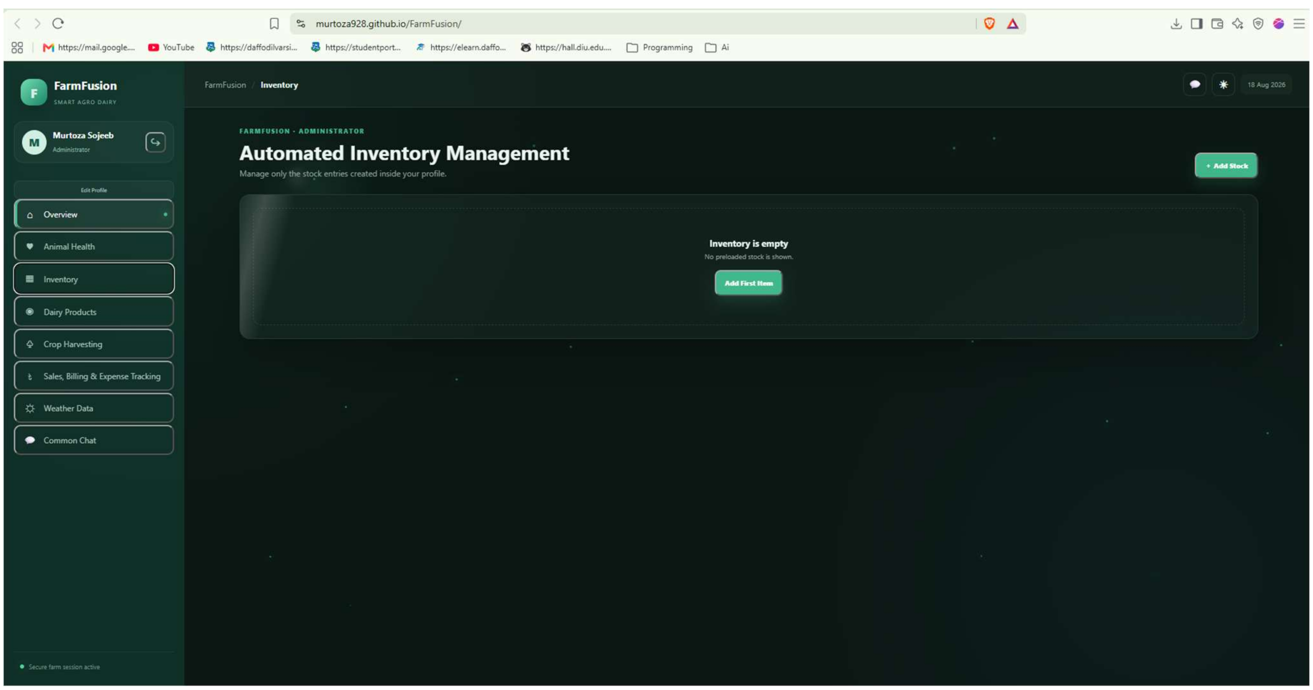

Figure 3.4.4: Inventory & Feeding Prototype

The Inventory & Feeding prototype emphasizes the information and actions most relevant to the selected role/module while keeping navigation consistent across the platform.

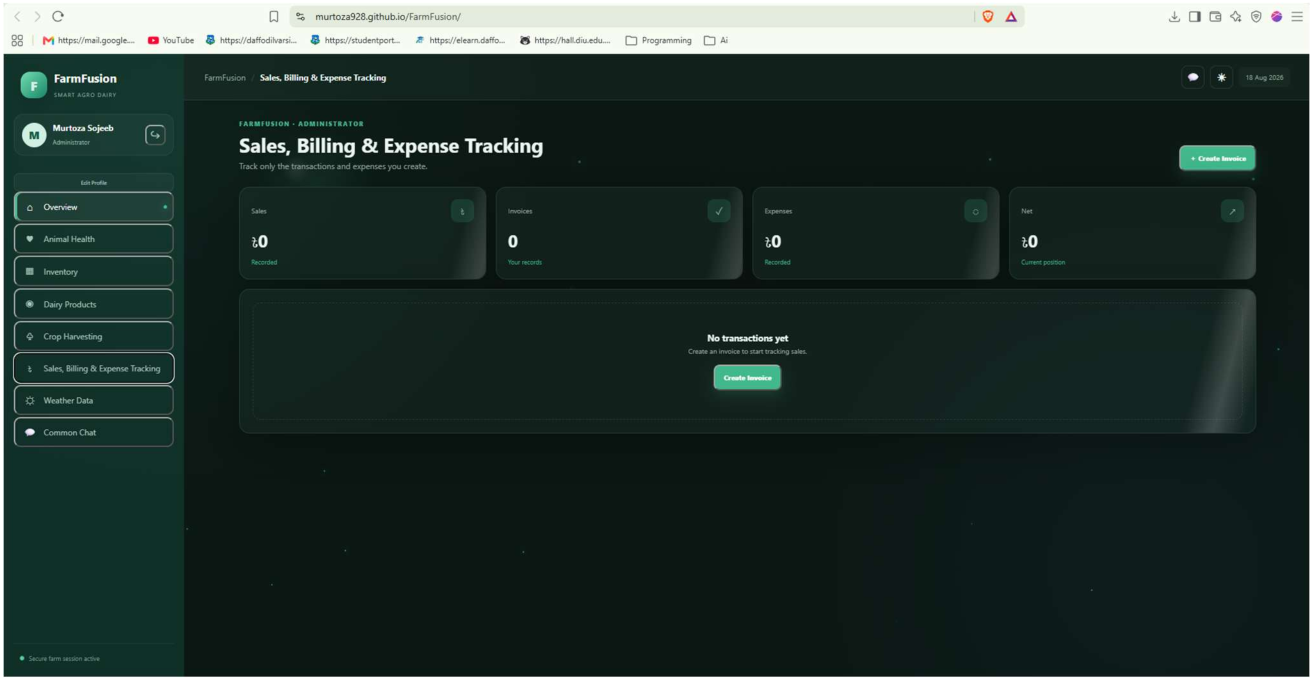

Figure 3.4.5: Sales, Billing & Expense Prototype

The Sales, Billing & Expense prototype emphasizes the information and actions most relevant to the selected role/module while keeping navigation consistent across the platform.

Chapter 4: Implementation & Testing

4.1 Module Implementation & Key Code Snippets

FarmFusion is organized into interconnected modules so that each operational function can be implemented and tested independently before integration.

User Registration & Login

Handles account creation, OTP/verification flow, credential validation, role checking and dashboard routing.

Animal Health Module

Handles livestock health status, health records, newborn movement, pregnancy monitoring and medicine/vaccine tracking.

Inventory Module

Handles item creation, quantity updates, reorder levels and low-stock alerts.

Feeding & Nutrition Module

Handles feed availability, meal scheduling, staff/food selection, feeding orders and confirmation.

Crop Harvesting Module

Handles crop/field selection, maturity, resources, harvest, yield, quality and dispatch.

Dairy Product Module

Handles product entries, damages, quantities, stock updates and confirmation.

Wholesale Module

Handles wholesale requests, availability, quotations, negotiation, confirmation/rejection and delivery reconciliation.

Sales, Billing & Expense Module

Handles sales orders, invoices, payments, expenses, verification and financial reporting.

Weather & Alert Module

Handles sensor/weather data, condition analysis, risk detection and notifications.

4.2 Database Implementation

The database implementation follows the entities and relationships defined in the ERD and class diagram. Core records include users, livestock, health records, inventory items, feeding sessions, crops, harvest records, dairy products, sales orders, invoices, payments, expenses, weather data and alerts. The data model is designed to support traceability from operational events to the users and records involved.

4.3 Test Suite Execution & Verification

Testing is organized into unit testing, integration testing and user-oriented acceptance testing. The test cases are derived from the success and alternative flows of the use case descriptions so that normal behavior and error conditions are both checked.

Test Case ID	Description	Expected Result	Status
TC-01	Registration with valid data	Verified account is created successfully	Pass
TC-02	Login with incorrect password	Authentication error is displayed and access is denied	Pass
TC-03	Animal health record with incomplete data	Validation prevents incomplete save	Pass
TC-04	Inventory quantity below reorder level	Low-stock alert is generated	Pass
TC-05	Feeding session confirmation	Feeding order and confirmation are stored	Pass
TC-06	Crop harvest before maturity	Harvest remains pending / user is prompted to reschedule	Pass
TC-07	Wholesale request with insufficient stock	Availability message is returned and order is not falsely confirmed	Pass
TC-08	Sales record with invalid amount	Validation/verification flags the discrepancy	Pass
TC-09	Severe weather condition detected	Alert is generated and delivered to responsible users	Pass

The test cases are intentionally aligned with the project's alternative flows. This keeps requirements, behavioral models and verification activities consistent and helps identify failures at module boundaries.

Chapter 5: Learning Outcome

5.1 Knowledge

- Understanding how to transform a real agricultural problem into functional and non-functional requirements.
- Understanding the different purposes of use case, activity, sequence, state, class, ER and DFD models.
- Understanding how data entities and relationships support operational workflows.
- Understanding how automated monitoring and alert generation can be integrated into a farm-management platform.
- Understanding the importance of consistency between requirements, diagrams, database design and test cases.

5.2 Skills

- Requirement analysis and structured use case writing.
- Creating readable UML and non-UML diagrams.
- Designing normalized data structures and mapping them to classes and entities.
- Writing workflow and interaction descriptions.
- Designing role-oriented interfaces and dashboard concepts.
- Testing both successful and alternative/error scenarios.
- Coordinating multiple modules into a single consistent software-engineering report.

5.3 Tools

- HTML, CSS and JavaScript for interface prototyping.
- Diagramming tools such as diagrams.net/Draw.io for system modeling.
- Microsoft Word-compatible document tools for report preparation.
- Image and diagram tools for preparing presentation-quality figures.
- Database/modeling tools appropriate for implementing the ERD and class model.

Chapter 6: Reflection on Project Proposal

6.1 How the Project Changed

The project evolved from a collection of individual farm-management functions into a more integrated smart agro ecosystem. During refinement, additional attention was given to sales, billing and expense tracking, more detailed animal-health monitoring, feeding workflows, wholesale operations, weather alerts and the relationships among the database entities.

The diagram set was also refined iteratively. Use cases were expanded into detailed descriptions, then translated into activity and sequence interactions, while the class and ER models were aligned with the operational data represented in the DFD. This iterative refinement is consistent with the project's selection of an Agile development approach.

6.2 Planned vs. Actual Timeline

Period	Planned Work	Observed/Completed Work
Weeks 1–2	Requirement gathering and scope definition	FarmFusion actors, modules, objectives and initial use cases were identified.
Weeks 3–4	System modeling	Use case, activity, sequence, state, class, ER and DFD models were refined.
Weeks 5–8	UI/UX and core module design	Dashboard/module concepts and operational workflows were developed.
Weeks 9–11	Integration and validation	Data flows, role permissions, alerts and cross-module relationships were checked.
Weeks 12–14	Testing and documentation	Test cases, report sections, figures and final presentation materials were prepared.

6.3 New Tools We Used

- Diagramming and modeling tools were used repeatedly to refine UML, ER and DFD consistency.
- Image-generation and editing workflows were used to prepare clearer presentation figures.
- Document automation was used to maintain consistent tables, captions, headings and page structure.
- Web research was used selectively to support the literature-review and quality-model sections.

6.4 Things That Did Not Go as Expected

- Keeping all diagrams consistent while adding new modules required repeated revisions.
- Some operational processes initially overlapped between inventory, feeding, dairy and finance, requiring clearer boundaries.
- Different diagram types use different modeling conventions, so the same workflow had to be expressed differently in activity, sequence and state views.
- Producing readable diagrams for a large system required simplifying labels and organizing information into multiple levels.

Chapter 7: Conclusion and Next Steps

7.1 Final Thoughts

FarmFusion – Smart Agro System provides a structured foundation for integrating major farm operations into one digital environment. The project combines role-based access with animal health, inventory, feeding, crop, dairy, wholesale, financial and weather-management functions. The complete model set demonstrates how the requirements can be transformed into architecture, data structures, workflows, interactions and verification activities.

The strongest aspect of the design is the relationship among the artifacts. The use cases define what the system must do; activity diagrams explain how workflows proceed; sequence diagrams show interaction order; state diagrams show lifecycle progression; the class and ER models define the data structure; and the DFD shows how operational data moves through the system.

7.2 What Can Be Done Next

- Integrate real IoT sensors and automated cameras for live farm monitoring.
- Add predictive analytics for animal health, crop yield, inventory demand and weather risk.
- Add stronger notification channels such as SMS, email and mobile push notifications.
- Add mobile applications for farmers, staff and veterinarians.

- Add advanced financial analytics and exportable accounting reports.
- Add multi-farm and multi-location support.
- Add stronger audit logging, backup, recovery and production-grade security controls.
- Conduct field testing with real farm users and refine the system based on operational feedback.

References

[1] Food and Agriculture Organization of the United Nations (FAO), "Smart Farming," FAO Smart Farming. Available at: https://www.fao.org/smart-farming/en

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[7] The FarmFusion project artifacts supplied during the system analysis and design process: use case, use case descriptions, activity, sequence, state, class, ER and DFD diagrams.

[8] Main Project Report PDF supplied as the formatting/reference model for this report.

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